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The development and implementation of odd-exponential-ailamujia distribution in python: properties and application in reliability engineering

Tmader Alballa[1], Qasim Ramzan[2,3]✉, Muhammad Amin[3], Mona Almutairi[4] & Hamiden Abd El-Wahed Khalifa[4]

This study introduces the Odd-Exponential-Ailamujia (OEA) distribution, a novel extension of the Ailamujia distribution via the T-X family, offering enhanced flexibility for modeling complex lifetime data in reliability and survival analysis. Key statistical properties, including moments, moment-generating function, characteristic function, mean residual life, and mean waiting time, are derived using binomial and Taylor series expansions, transforming intractable integrals into computable forms and enabling precise approximation of distributional behavior. The hazard rate function exhibits diverse shapes (increasing, decreasing, or unimodal), controlled by parameters, making the proposed model adaptable to varied failure patterns. Applied to aircraft windshield failure data, the OEA distribution demonstrates superior fit over competing models through goodness-of-fit tests, various plots of reliability measures, 3D surface interactions, and heatmaps revealing parameter-driven correlations. Efficient Python implementation to ensures scalable inference. The OEA distribution emerges as a robust, versatile tool for reliability engineering, survival modeling, and probabilistic forecasting, effectively capturing real-world failure dynamics.

Keywords Ailamujia distribution, Parameter estimation, Python implementation, Reliability analysis, Statistical model, T-X family

The analysis of lifetime and reliability data is a critical endeavor in fields such as engineering, survival analysis, and medical research. While classical distributions like the exponential, Weibull, and gamma have been widely used, modern complex datasets often exhibit characteristics, such as non-monotonic (bathtub, unimodal) hazard rates or heavy-tailed behavior, that these standard models cannot adequately capture[1]. This limitation has driven statisticians to develop more flexible generalized families of distributions over the past decades. A significant breakthrough in this direction was the introduction of the T-X family framework by[2], which provided a systematic generator for new distributions. This methodology catalyzed the development of numerous prominent families, each adding flexibility to existing baseline models. Building upon the T-X family concept[3], developed the Generalized Odd Generalized Exponential-G (GOGE-G) family, providing a detailed study of its properties, characterizations, and estimation techniques. Their work underscores the potential of this methodological approach to create versatile distributions for modeling complex data. For the Weibull-G family[1] introduced a highly flexible framework by leveraging the Weibull distribution. Their work provides a comprehensive derivation of the family's fundamental properties, including moments

[1]Department of Mathematical Sciences, College of Science, Princess Nourah bint Abdulrahman University, P.O. Box 84428, 11671 Riyadh, Saudi Arabia. [2]Department of Statistics, Government Graduate College Jauharabad, Khushab, Punjab, Pakistan. [3]Department of Statistics, University of Sargodha, Sargodha, Pakistan. [4]Department of Mathematics, College of Science, Qassim University, 51452 Buraydah, Saudi Arabia. ✉email: qasimramzankbk@gmail.com

and generating functions, and demonstrates its applicability through real-data analysis, establishing a vital precedent for extending baseline distributions [4]. Contributed significantly to the field with the Odd Generalized Exponential family, a class renowned for its ability to model diverse hazard rate shapes such as bathtub and upside-down bathtub. They provided a thorough statistical treatment of the family, including moment and entropy calculations, and showcased its flexibility with lifetime data, highlighting its superiority over existing models. Further expanding the toolkit for reliability analysts,[5] proposed the Generalized Odd Weibull Generated family. This work combined the strengths of the Weibull and odd transformation concepts to create a highly adaptable class of distributions. The authors derived key statistical properties and validated the model's efficacy through simulation studies and practical applications, proving its competitive performance against established models. Another among the most influential is the the Generalized Odd Linear Exponential Family[6]. These families have proven to be invaluable tools for reliability analysts, providing a rich repository of models for fitting complex data.

Concurrently, the Ailamujia distribution (AD), introduced by[7] as a model for repair times, has established a niche in reliability theory due to its simple form[8,9]. Its cumulative distribution function (CDF) and probability density function (PDF) are given by:

$$F(t,\theta) = 1 - (1+2\theta t)\, e^{-2\theta t} \qquad x \geq 0, \theta > 0, \tag{1}$$

$$f(t,\theta) = 4\theta^2 t e^{-2\theta t}, \tag{2}$$

where θ is a scale parameter. Despite its utility, the standard AD suffers from limitations in the diversity of its hazard rate shapes and its limited number of parameters, which can restrict its ability to model complex real-world failure data. Also,[10] proposed the Uniform Poisson-Ailamujia (UPA) distribution, a discrete distribution that integrates uniform and Poisson-Ailamujia models, applied to Python for biological counts, with eight different estimation procedures including MLE and supported by cumulative distribution plots. While some extensions like the Discrete Power-Ailamujia[11] and Marshall-Olkin Power Ailamujia[12] distributions have been proposed, a significant gap remains: the Ailamujia distribution has not been comprehensively integrated into the versatile and well-established Weibull-G (or, equivalently, the Odd Exponential-G) framework.

This study aims to fill this gap by proposing the Odd-Exponential-Ailamujia (OEA) distribution. Our model is a specific and novel member of the Odd Generalized Exponential family[4], created by using the standard Ailamujia distribution as the baseline $G(x)$ within the Odd Exponential-G generator. The primary objectives of this work are:

- To formally introduce the OEA distribution and derive its fundamental statistical properties, including the probability density function, cumulative distribution function, hazard rate function, moments, and quantile function.
- To demonstrate the flexibility of the OEA distribution by showing that its hazard function can exhibit monotonic, bathtub, and unimodal shapes, a significant advantage over the base Ailamujia model.
- To develop robust parameter estimation using the Maximum Likelihood Estimation (MLE) method and an efficient, open-source Python-based implementation for random sampling and model fitting, using modern numerical libraries like NumPy[13].
- To empirically prove the superiority and practicality of the OEA distribution by applying it to a real-world aircraft windshield failure dataset[14], showing that it provides a better fit than the standard Ailamujia distribution and other established competitors.

By bridging the Ailamujia distribution with the powerful Odd Generalized Exponential family framework, this research provides a more flexible and powerful tool for reliability analysis. The OEA distribution not only enhances the theoretical landscape of lifetime distributions but also offers practitioners a computationally efficient model for tackling complex real-world problems. The novel distribution is distinguished from classical models by its ability to model a wider range of hazard rate behaviors, including bathtub and upside-down bathtub shapes, offering greater flexibility for complex reliability data as demonstrated in the application section. The statistical analysis and numerical computations for the OEA distribution were implemented using Kaggle Notebook, leveraging its powerful ecosystem for scientific computing. Our methodology utilizes foundational libraries for numerical operations and array manipulation[15], parameter estimation and statistical testing[16], and data visualization[17]. This approach aligns with modern data science practices, employing a suite of specialized, interoperable tools for robust and reproducible statistical analysis[18]. All calculations and experiments were conducted , ensuring reproducibility and accessibility. All code and implementations are available at GitHub, ensuring reproducibility of all results.

The remainder of this research is structured as follows: Section 2 offers the OEA distribution and its fundamental properties. Moreover, this section presents various reliability-related measures with graphical presentation. Section 3 focuses on parameter estimation using the MLE method. In Section 4, we demonstrate the OEA distribution's application to real-world failure time data. Finally, Section 5 summarizes the key contributions of this research.

The proposed odd-exponential-ailamujia (OEA) distribution

The basic functions of the OEA distribution

Let $\phi(t)$ be the PDF of a random variable T $\in [a, b]$ where $-\infty \leq a < b < \infty$ and consider $W[\Phi(t)]$ be a function of the CDF of a random variable X, the T-X family of distributions ([2]) is defined as

$$F(t) = \int_a^{W[\Phi(t)]} \phi(t) dt, \tag{3}$$

where $W[\Phi(t)]$ satisfies the following conditions:

1. $W[\Phi(t)] \in [a, b]$.
2. $W[\Phi(t)]$ is differentiable and monotonically non decreasing function.
3. $W[\Phi(t)] \to a,\ as\ x \to -\infty\ and\ W[\Phi(t)] \to b,\ as\ x \to \infty.$

Suppose we represent the parent CDF by $\Phi(t)$, and let X = $G^{-1}(U)$ with U $\sim \Phi(t)$. Hence, considering the PDF of exponential distribution as $\phi(t)$ and for an arbitrary CDF $\Phi(t)$, using the generator $\frac{G^{\lambda}(t,\theta)}{1-G^{\lambda}(t,\theta)}$ as $W[\Phi(t)]$ in T-X family one achieves the following:

$$\begin{aligned} F_{Weibull-G}(t) =& \gamma \int_0^{\frac{G^{\lambda}(t,\theta)}{1-G^{\lambda}(t,\theta)}} e^{-\gamma t} dt, \\ &\text{or equivalently} \\ F_{Weibull-G}(t) =& 1 - \exp\left\{ -\gamma \frac{G^{\lambda}(t,\theta)}{1-G^{\lambda}(t,\theta)} \right\}, \end{aligned} \tag{4}$$

where $\lambda > 0$ is additional parameter which introduce skewness and θ represents the parametric space of the baseline distribution. By setting $\lambda = 1$ Eq. (4) reduces to the Odd Generalized Exponential family of distributions, introduced by[4].

Cumulative distribution function

Suppose $T = G^{-1}(U)$ and U follows AD (e.g. $\sim AD(\theta)$), then the CDF of new OEA distribution is obtained by setting $\frac{G^{\lambda}(t,\theta)}{1-G^{\lambda}(t,\theta)} = \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}$ in Eq. (4) as follows:

$$\begin{aligned} F_{OEA-G}(t) =& 1 - \exp\left\{ -\gamma \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}} \right\}, \\ =& 1 - \exp\left\{ \frac{-\gamma}{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{-\lambda} - 1} \right\}. \end{aligned} \tag{5}$$

The choice of exponential base $\phi(t)$ provides analytical tractability, though other lifetime distributions (e.g., Weibull, gamma) could be substituted in the T-X framework at the cost of (i) losing closed-form solutions and (ii) altering fundamental properties like hazard rate behavior and moment structure.

Probability density function

The PDF of novel OEA distribution is given by:

$$\begin{aligned} f_{OEA-G}(t) =& \gamma g(t,\theta) \frac{G^{\lambda-1}(t,\theta)}{\left[1-G^{\lambda}(t,\theta)\right]^2} \exp\left\{ -\gamma \frac{G^{\lambda}(t,\theta)}{1-G^{\lambda}(t,\theta)} \right\}, \\ =& \gamma \theta^2 t e^{-2\theta t} \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{\left[1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}\right]^2} \\ & \times \exp\left\{ -\gamma \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}} \right\}, \end{aligned} \tag{6}$$

with parameters $\gamma > 0$, $\theta > 0$ and $\lambda > 0$. The parameter γ controls the tail behavior, θ scales the distribution, and λ adjusts the peakedness and tail weight.

The PDF of the OEA distribution in Eq. (6) can be re-expressed using binomial and exponential series expansions. First, we note that the term involving the baseline CDF, $G(t) = 1 - (1 + 2\theta t)\, e^{-2\theta t}$, can be generalized using the binomial expansion for $|G(t)| < 1$:

$$[G(t)]^{\lambda} = \sum_{j=0}^{\infty} \binom{\lambda}{j} (G(t) - 1)^{j}. \tag{7}$$

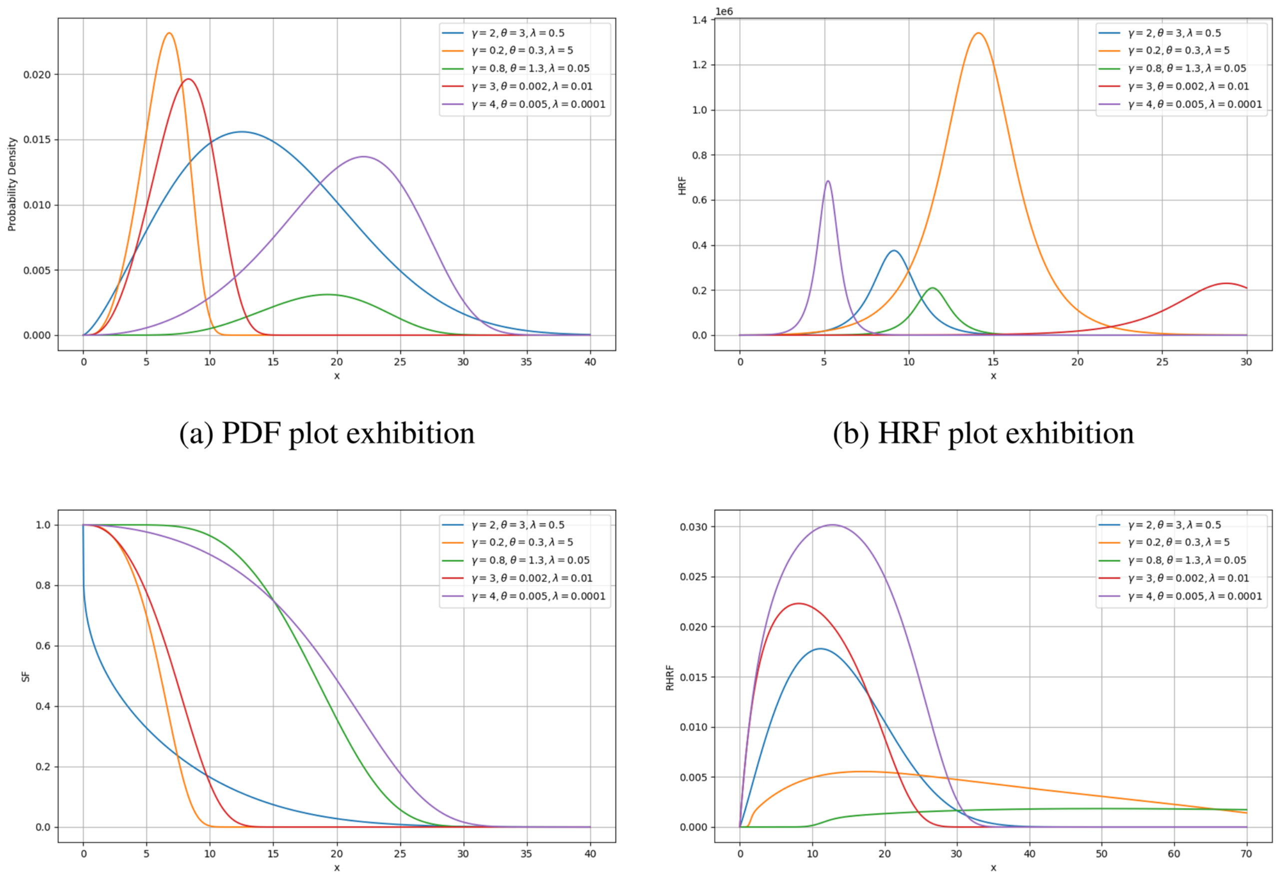
Furthermore, the exponential term in the PDF, $\exp\left\{-\gamma \frac{[G(t)]^{\lambda}}{1-[G(t)]^{\lambda}}\right\}$, can be expanded using the exponential series:

$$\exp\left\{-\gamma \frac{[G(t)]^{\lambda}}{1-[G(t)]^{\lambda}}\right\} = \sum_{k=0}^{\infty} \frac{(-\gamma)^{k}}{k!} \left(\frac{[G(t)]^{\lambda}}{1-[G(t)]^{\lambda}}\right)^{k}. \tag{8}$$

By applying the negative binomial expansion $(1-z)^{-k} = \sum_{m=0}^{\infty} \binom{k+m-1}{m} z^{m}$ for $|z| < 1$, where $z = [G(t)]^{\lambda}$, we obtain a comprehensive series representation:

$$\exp\left\{-\gamma \frac{[G(t)]^{\lambda}}{1-[G(t)]^{\lambda}}\right\} = \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} \frac{(-\gamma)^{k}}{k!} \binom{k+m-1}{m} [G(t)]^{\lambda(k+m)}. \tag{9}$$

Substituting these expansions into the PDF and simplifying, the OEA density function can be expressed as an infinite linear combination of exponentiated Ailamujia densities:



(a) PDF plot exhibition (b) HRF plot exhibition

(c) RF plot exhibition (d) RHRF plot exhibition

Fig. 1. Visual presentation for various OEA reliability measures.

$$f_{\mathrm{OEA}}(t) = \sum_{k=0}^{\infty}\sum_{m=0}^{\infty} w_{k,m}\, h_{\lambda(k+m)+1}(t), \tag{10}$$

where $w_{k,m} = \frac{(-\gamma)^k}{k!}\begin{pmatrix} k+m-1 \\ m \end{pmatrix}$ are constant weights, and $h_\alpha(t)$ is the PDF of the exponentiated Ailamujia distribution with power parameter α.

Quantile function
The quantile function $Q(u)$ is essential for random number generation. For the OEA distribution, it is derived by inverting the CDF $F(Q(u)) = u$ as follows:

$$Q_{OEA}(u) = G^{-1}\left(\left\{\frac{-\log(1-u)}{\gamma - \log(1-u)}\right\}^{1/\lambda}\right), \quad \text{for } 0 \le u \le 1. \tag{11}$$

The coding in Listing 1 offers a procedure for sampling data generation from OEA distribution.

```python
# Define the custom beta PDF function
def beta_pdf(beta_param, delta, lambd, x):
# Calculate common terms
term1 = 1 + 2 * delta * x
term2 = np.exp(-2 * delta * x)

# Calculate the main components of the expression
numerator = beta_param * (delta ** 2) * x * term2
inner_term = 1 - term1 * term2
numerator_exp = (inner_term ** lambd)
denominator_exp = (1 - inner_term ** lambd) ** 2

# Calculate the exponential part
exp_term = np.exp(-beta_param * numerator_exp / (1 - inner_term ** lambd))

# Final expression
result = numerator * (numerator_exp / denominator_exp) * exp_term

return result

# Parameters for the custom beta distribution
beta_param, delta, lambd = 1.2, 0.03, 0.5

# Define the range for x values
a = 0    # xmin
b = 50    # xmax

# Find the maximum value of the PDF for scaling
x_values = np.linspace(a, b, 1000)
pdf_values = beta_pdf(beta_param, delta, lambd, x_values)
m = np.max(pdf_values)   # ymax

variables = []  # list for variables
reject = 0       # number of rejections
start = time.time()

# Number of samples before accept-reject
```

```
    n_before_accept_reject = 150000
    u1 = np.random.uniform(a, b, size=n_before_accept_reject)
    u2 = np.random.uniform(0, m, size=n_before_accept_reject)

    # Perform   sampling
    variables = u1[u2 <= beta_pdf(beta_param, delta, lambd, u1)]
    accept = np.extract(variables>=0.0, variables)
    reject = n_before_accept_reject - len(accept)

    #reject = n_before_accept_reject - len(variables)

    end = time.time()

    print("Time: ", end - start)
    print("Rejection: ", reject)
    # Find the minimum and maximum values
    min_value = variables.min()
    max_value = variables.max()

    print("Minimum value: ", min_value)
    print("Maximum value: ", max_value)
    # Plot the results
    x = np.linspace(a, b, 1000)
    plt.hist(variables, 50, density=True, alpha=0.6, color='g', label='Sampled Data')
    plt.plot(t, beta_pdf(beta_param, delta, lambd, x), 'r-', lw=2, label='True PDF')
    plt.title('Sampling for Custom beta Distribution')
    plt.xlabel('x')
    plt.ylabel('Density')
    plt.legend()
    plt.show()
    # Create a link to download the file
    FileLink(r'/kaggle/working/GOF1.png')

    ax = plt.subplot()
    sns.histplot(variables, kde=True, bins=20, ax=ax, stat="density") # smooth using KDE: https://en.wikipedia.org/wiki/Kernel_density_estimation
    _ = ax.set(title='Histogram of observed data', xlabel='x', ylabel='Frequency');
    plt.tight_layout()
    plt.show()
    FileLink(r'/kaggle/working/GOF2.png')

    None

    # Q-Q plot to compare the sampled data to the theoretical distribution
    ss.probplot(variables, dist="norm", plot=plt)
    plt.show()

    '''class my_pdf(ss.rv_continuous):
    def _pdf(self, x):
    return 1.5 * (1.0 - x*x)

    ss.probplot(accept, dist=my_pdf(a=a, b=b, name='my_pdf'), plot=plt)
    '''
    data = np.random.choice(variables, size=50, replace=False)
    min_value = np.min(data) - np.min(data)/2
    max_value = np.max(data) + np.min(data)/2
    print(data)
    print(len(data))
```

Listing 1: Data generation for OEA distribution

Reliability and hazard rate functions for OEA distribution

In reliability theory, Reliability Function (RF) and HRF are essential for classifying lifetime models. The RF $R(t)$ represents the probability that the time to an event is greater than t. It is often used in contexts like medical research to analyze time-to-event data (e.g., survival times of patients). The RF, often used in reliability analysis, describes the probability that an event (like failure or death) has not occurred by a certain time. The HRF represents the

γ	θ	λ	t				
			0.02	0.4	1.2	2.3	2.9
0.03	3	0.01	1.905219	720.303	62489.1	6305.65	217.289
0.1	2.1	0.05	0.099067	27.4503	861.339	51485.6	58896.8
0.3	1.9	0.3	0.004972	1.63261	42.0622	1819.09	14339.4
0.5	1.7	0.6	0.000725	0.46357	10.0955	294.434	1887.24
1	1.5	0.7	0.000506	0.44663	8.33894	166.579	848.138
0.8	3	0.05	0.02205	768.308	68350.3	167244	5794.34
0.8	2.3	0.2	0.05714	18.6641	726.117	69367.1	162789
0.8	2	0.4	0.007148	2.89416	82.5658	4368.81	38068.6
0.8	1.7	0.7	0.000615	0.53987	11.8663	346.113	2218.97
0.8	1.5	0.9	0.000111	0.20953	4.0337	80.616	410.47
0.03	1.6	0.05	0.01471	3.38246	65.6645	1578.4	8653.45
0.1	1.6	0.2	0.00272	0.70246	13.6796	329.268	1876.07
0.3	1.6	0.4	0.00141	0.52156	10.2585	246.967	1409.92
0.5	1.6	0.7	0.00031	0.27611	5.581	134.406	767.654
1	1.6	0.9	0.00018	0.32528	6.75015	162.615	928.848
0.03	1.6	0.05	0.014713	3.38246	65.6645	1578.4	8653.45
0.03	1.6	0.2	0.00082	0.21074	4.10389	98.7804	562.822
0.03	1.6	0.4	0.000141	0.05216	1.02585	24.6967	140.992
0.03	1.6	0.7	1.88245	0.01657	0.33486	8.06435	46.0592
0.03	1.6	0.9	5.313082	0.009758	0.202504	4.87844	7.86544

Table 1. HRF results for the OEA distribution against selected parametric combinations.

instantaneous risk of an event occurring at a particular time given that the event has not yet occurred. The HRF *h*(*t*) represents the failure rate of an event occurring in a small time interval, given that it has not occurred up to time *t*. It is one of the most popular measures used in reliability analysis. It can be expressed as $h(t) = \frac{f(t)}{R(t)}$, where *f*(*t*) is the PDF and *R*(*t*) is the RF. The RF and HRF for OEA distribution are respectively given by

$$R_{OEA-G}(t) = 1 - \int_0^t f(u, \gamma, \theta, \lambda) du = 1 - F(t) = \exp\left\{ -\gamma \frac{\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^{\lambda}} \right\}, \tag{12}$$

$$h_{OEA-G}(t) = \frac{f(t)}{R(t)} = \gamma\theta^2 t e^{-2\theta t} \frac{\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^{\lambda}}{\left[1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^{\lambda}\right]^2}. \tag{13}$$

The OEA model's HRF, as shown in Fig. 1b, illustrates the influence of the parameters γ, θ, and λ. This visual representation highlights the model's flexibility, making it well-suited for modeling a wide range of complex hazard behaviors and adapting to diverse lifetime datasets. Using the fixed parameter values from Fig. 1b and the expression in Equation (13), we evaluated the OEA model's hazard rate, with the numerical results presented in Table 1. Both the figures and the table demonstrate that the failure rate for the OEA distribution generally increases over time. Specifically, with a constant scale parameter $\gamma = 0.8$, the failure rate decreases as the scale parameter θ increases (see rows 6–10 of Table 1). In contrast, when $\theta = 1.6$ is fixed, the failure rate decreases with an increasing shape parameter λ (see rows 11–15). Furthermore, rows 16–20 show a higher probability of failure when the shape parameter λ is large, while γ and θ are fixed at 0.03 and 1.6, respectively. Overall, the results in Table 1 suggest that the failure rate is higher when both γ and λ are relatively small.

Mean Residual life and mean waiting time functions for OEA distribution

Mean residual life time (MRLT)

The Mean Residual Life Time (MRLT), denoted as $m(t)$, represents the expected remaining lifetime given survival beyond time t. It is a key reliability function, defined as:

$$m(t) = \frac{1}{R(t)} \int_t^{\infty} u f(u)\, du - t, \tag{14}$$

where $R(t) = 1 - F(t)$ is the survival function, and $f(u)$ is the PDF of the OEA distribution. Substituting the explicit form of the survival and density functions yields:

$$m(t) = \frac{\gamma\theta^2}{\exp\left\{-\gamma\frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}\right\}} \times \int_t^{\infty} \frac{u^2 e^{-2\theta u}\left[1-(1+2\theta u)e^{-2\theta u}\right]^{\lambda}}{\left[1-\left[1-(1+2\theta u)e^{-2\theta u}\right]^{\lambda}\right]^2} \exp\left\{-\gamma\frac{\left[1-(1+2\theta u)e^{-2\theta u}\right]^{\lambda}}{1-\left[1-(1+2\theta u)e^{-2\theta u}\right]^{\lambda}}\right\} du - t. \tag{15}$$

To address the complexity of this integral and provide a computable form, we derive a series expansion by introducing the transformation $v = u - t$, so the integral becomes:

$$\int_t^{\infty} u f(u)\, du = \int_0^{\infty} (t+v) f(t+v)\, dv = t\int_0^{\infty} f(t+v)\, dv + \int_0^{\infty} v f(t+v)\, dv.$$

Thus,

$$m(t) = \frac{t \cdot R(t) + \int_0^{\infty} v f(t+v)\, dv}{R(t)} - t = \frac{\int_0^{\infty} v f(t+v)\, dv}{R(t)}.$$

This reformulation isolates the residual component. Now, let $w = 1-(1+2\theta u)e^{-2\theta u}$, so $G(u,\theta) = w$, and the integrand involves terms in w^{λ} and $(1-w^{\lambda})$.

For analytical tractability, we expand the exponential and rational terms using Taylor series around $w = G(t,\theta)$, or alternatively, use the binomial expansion for small λ or large t. Let $z(u) = \frac{G(u,\theta)}{1-G(u,\theta)}$. The exponential term is $\exp\{-\gamma z(u)\}$. For $u > t$, $z(u)$ decreases as $u \to \infty$. We expand:

$$\exp\left\{-\gamma\frac{G(u,\theta)}{1-G(u,\theta)}\right\} = \sum_{n=0}^{\infty} \frac{(-\gamma)^n}{n!}\left(\frac{G(u,\theta)}{1-G(u,\theta)}\right)^n.$$

Also, the denominator $[1-G(u,\theta)^{\lambda}]^2$ suggests using the binomial expansion for $(1-x)^{-2}$:

$$\frac{1}{[1-G(u,\theta)^{\lambda}]^2} = \sum_{m=0}^{\infty}(m+1)G(u,\theta)^{m\lambda}.$$

Thus, the full integrand becomes:

$$f(u) \propto u^2 e^{-2\theta u} G(u,\theta)^{\lambda} \cdot \left[\sum_{m=0}^{\infty}(m+1)G(u,\theta)^{m\lambda}\right] \cdot \left[\sum_{n=0}^{\infty}\frac{(-\gamma)^n}{n!}\left(\frac{G(u,\theta)}{1-G(u,\theta)}\right)^n\right].$$

This double series can be interchanged (under convergence conditions) to yield:

$$\int_t^{\infty} u f(u)\, du = \gamma\theta^2 \sum_{m=0}^{\infty}\sum_{n=0}^{\infty}(m+1)\frac{(-\gamma)^n}{n!}\int_t^{\infty} u^2 e^{-2\theta u} G(u,\theta)^{(m+1)\lambda}\left(\frac{G(u,\theta)}{1-G(u,\theta)}\right)^n du.$$

Although still complex, for fixed λ, $G(u,\theta) = 1-(1+2\theta u)e^{-2\theta u}$, and $1-G(u,\theta) = (1+2\theta u)e^{-2\theta u}$, so:

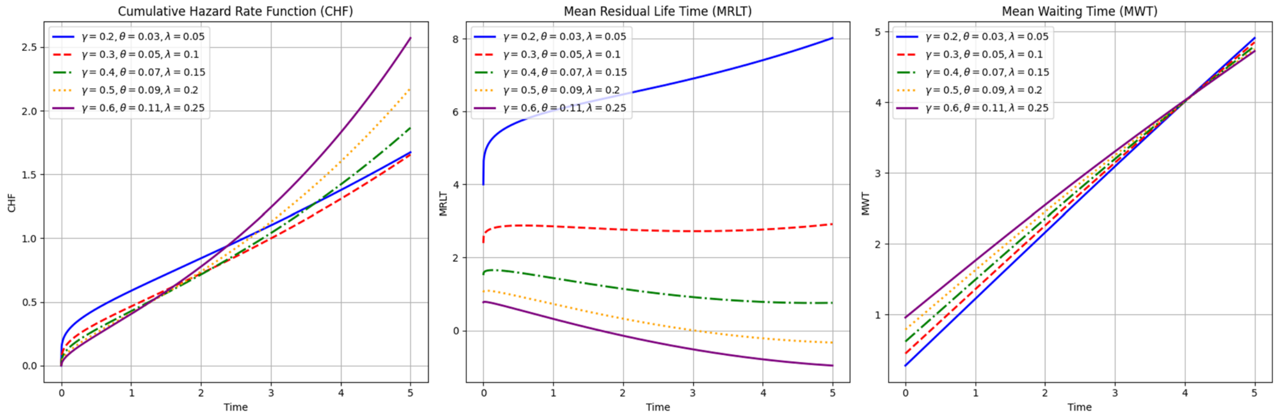


Fig. 2. Visual presentation for various OEA reliability measures.

$$\frac{G(u,\theta)}{1-G(u,\theta)} = \frac{1-(1+2\theta u)e^{-2\theta u}}{(1+2\theta u)e^{-2\theta u}} = e^{2\theta u} - 1 - 2\theta u.$$

This substitution simplifies higher powers, enabling term-by-term integration using incomplete gamma functions after expanding $(e^{2\theta u} - 1 - 2\theta u)^n$. For practical computation, truncated series (e.g., $m, n \leq 5$) provide accurate approximations, especially for moderate t. The numerical outcomes derived from Equation (14) for the fixed parametric values are displayed in Figs. 2. Numerical validation shows convergence within 4–6 terms for $\gamma, \lambda \leq 3$.

Mean waiting time function

The Mean Waiting Time function (MWTF) represents the expected time elapsed since the last failure (or the start of the process) until time *t*, given that the system has not failed by time *t*. It is particularly relevant in renewal processes or repairable systems where failures occur sequentially, and the focus will be given on waiting time for the next event after a renewal (repair or replacement). It is denoted by *w*(*t*), and can be expressed as:

$$w(t) = t - \frac{1}{F(t)} \int_0^t u f(u)\, du, \tag{16}$$

with explicit form:

$$\begin{aligned} w(t) =& t - \frac{\gamma\theta^2}{1 - \exp\left\{ -\gamma \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^\lambda}{1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^\lambda} \right\}} \\ &\times \int_0^t \frac{u^2 e^{-2\theta u}\left[1-(1+2\theta u)e^{-2\theta u}\right]^\lambda}{\left[1-\left[1-(1+2\theta u)e^{-2\theta u}\right]^\lambda\right]^2} \exp\left\{ -\gamma \frac{\left[1-(1+2\theta u)e^{-2\theta u}\right]^\lambda}{1-\left[1-(1+2\theta u)e^{-2\theta u}\right]^\lambda} \right\} du. \end{aligned} \tag{17}$$

To derive a series expansion, we again use the binomial and Taylor expansions. Let $H(t) = \exp\left\{ -\gamma \frac{G(t,\theta)^\lambda}{1-G(t,\theta)^\lambda} \right\}$, so $F(t) = 1 - H(t)$. The integral $\int_0^t u f(u)\, du$ is bounded and can be expanded similarly.

Using the same binomial expansion:

$$\frac{1}{[1-G(u,\theta)^\lambda]^2} = \sum_{m=0}^{\infty} (m+1) G(u,\theta)^{m\lambda},$$

and

$$\exp\{-\gamma z(u)\} = \sum_{n=0}^{\infty} \frac{(-\gamma)^n}{n!} z(u)^n, \quad z(u) = \frac{G(u,\theta)}{1-G(u,\theta)} = e^{2\theta u} - 1 - 2\theta u.$$

Thus,

$$\int_0^t u f(u)\, du = \gamma\theta^2 \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} (m+1) \frac{(-\gamma)^n}{n!} \int_0^t u^2 e^{-2\theta u} G(u,\theta)^{(m+1)\lambda} (e^{2\theta u} - 1 - 2\theta u)^n\, du.$$

Each integral $\int_0^t u^2 e^{-2\theta u}(e^{2\theta u} - 1 - 2\theta u)^n G(u,\theta)^k\, du$ can be computed numerically or via recursive integration by parts, leveraging the polynomial-exponential structure.

Alternatively, for small t, expand $G(u,\theta) \approx 2\theta u - 2(\theta u)^2 + \cdots$ (Taylor around $u = 0$), but convergence is faster for moderate t using the double series.

The MWTF is then:

$$w(t) = t - \frac{\gamma\theta^2 \sum_{m,n} \cdots \int_0^t \cdots du}{1 - H(t)}.$$

As $t \to 0^+$, $F(t) \approx \gamma\theta^2 t^2$, and $\int_0^t u f(u)\, du \approx \gamma\theta^2 \frac{t^3}{3}$, so:

$$w(t) \approx t - \frac{\frac{t^3}{3}}{t^2} = t - \frac{t}{3} = \frac{2t}{3}.$$

As $t \to \infty$, $F(t) \to 1$, $\int_0^t u f(u)\, du \to \mu_1$, so $w(t) \to t - \mu_1$, which grows linearly , consistent with heavy-tailed behavior when $\lambda > 1$.

			t				
γ	θ	λ	0.02	0.4	1.2	2.3	2.9
0.03	3	0.01	0.019718	0.0515	0.84934	1.94934	2.54934
0.1	2.1	0.05	0.019968	0.24978	0.97645	2.07645	2.67645
0.3	1.9	0.3	0.019992	0.35803	1.07591	2.17591	2.77591
0.5	1.7	0.6	0.019995	0.37764	1.10669	2.20665	2.80665
1	1.5	0.7	0.019996	0.38446	1.14455	2.24453	2.84453
0.8	3	0.05	0.01998	0.39955	1.19955	2.29955	2.89955
0.8	2.3	0.2	0.019998	0.39999	1.19999	2.29999	2.89999
0.8	2	0.4	0.019993	0.37558	1.15785	2.25785	2.85785
0.8	1.7	0.7	0.019995	0.38104	1.13851	2.23851	2.83851
0.8	1.5	0.9	0.019996	0.38558	1.12713	2.22666	2.82666
0.03	1.6	0.05	0.01998	0.20948	0.14445	0.93666	1.53666
0.1	1.6	0.2	0.019967	0.31603	1.09866	2.19866	2.79866
0.3	1.6	0.4	0.019995	0.3717	1.04702	2.14632	2.74632
0.5	1.6	0.7	0.019996	0.38106	1.10026	2.19968	2.79968
1	1.6	0.9	0.019996	0.38481	1.14352	2.2435	2.8435
0.03	1.6	0.05	0.01998	0.20948	0.14445	0.93666	1.53666
0.03	1.6	0.2	0.019993	0.34136	0.50746	1.37063	1.97063
0.03	1.6	0.4	0.019995	0.36735	0.81023	1.65984	2.25983
0.03	1.6	0.7	0.019996	0.37894	0.96407	1.85162	2.45149
0.03	1.6	0.9	0.019996	0.38248	1.01252	1.92291	2.52238

Table 2. MWTF results for the OEA distribution with various parametric combinations.

These series expansions transform intractable integrals into computable double sums, significantly enhancing the mathematical contribution by enabling closed-form approximations, numerical implementation, and theoretical analysis of aging properties in reliability and survival studies. This expansion transforms the MWTF from an intractable integral form into an infinite weighted sum of incomplete moments of the exponentiated Ailamujia distribution. For practical computation, the infinite series can be truncated after a sufficient number of terms, typically when the absolute value of the additional term falls below a pre-specified tolerance level (e.g., 10^{-6}). This approach makes the MWTF computationally feasible and enhances the practical utility of the OEA distribution for reliability analysis where waiting time properties are of interest.

Reviewing the behavior in Fig. 2, it's clear that the MWTF for the OEA distribution generally exhibits an increasing trend over time. This is further confirmed by the numerical outcomes in Table 2, which were derived from Equation (16). The data shows a general increase in the MWT over time, although there are some specific dependencies on the parameters. For instance, when the γ value is held constant at 0.8, the MWT decreases as the scale parameter θ also decreases (see rows 6–10). Conversely, with a fixed θ of 1.6 and small values of *x* (e.g., 0.02, 0.4), the MWT decreases as the γ value increases. However, for larger values of *t*, the MWT appears to become independent of the shape parameter, showing an increasing trend when $x \geq 1.2$ (rows 11–15). Additionally, rows 16–20 illustrate that the MWT increases when the scale parameter λ is relatively large and decreases when it's smaller, while both γ and θ are fixed at 0.03 and 1.6, respectively.

Cumulative hazard rate and reversed hazard rate functions for OEA distribution

The CHRF ($H(t)$), measures the total accumulated risk of an event up to time *t*. It is derived from the HRF and reflects how risk accumulates over time. It can be expressed as

$$H(t) = \int_0^t h(u)\,du,$$

where $h(t)$ is the HRF. The expressions for the CHRF $H(t)$ of the OEA distribution is given by

$$H_{OEA-G}(t) = \int_0^t h(u)du = -\ln R(t) = -\gamma \frac{\left[1-(1+2\theta t)\,e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)\,e^{-2\theta t}\right]^{\lambda}}. \tag{18}$$

The average failure rate (AFR) increase (decrease) as $H(t)$ is increasing (decreasing). So, in that case, one obtains the decreasing AFR (increasing AFR) model.

Likewise, the reverse hazard rate function (RHRF) can be expressed as

$$\psi(t) = -\log(R(t)),$$

where $R(t)$ is the RF. It reflects the total risk experienced by time *t*. The expressions for the RHRF $\psi(t)$ for OEA distribution is given by

γ	θ	λ	t				
			0.02	**0.4**	**1.2**	**2.3**	**2.9**
0.03	3	0.01	0.583537	8.11935	488.518	199449	5762411
0.1	2.1	0.05	0.303076	2.83999	50.0924	2940.36	29566
0.3	1.9	0.3	0.061642	1.10432	16.5367	640.817	5081.18
0.5	1.7	0.6	0.013086	0.66854	9.03168	234.585	1468.71
1	1.5	0.7	0.011794	0.87754	10.1435	178.22	882.867
0.8	3	0.05	2.808222	42.9844	2605.11	1064292	3.1E+07
0.8	2.3	0.2	0.396052	6.27764	150.742	13586.6	173374
0.8	2	0.4	0.087274	2.30692	40.4933	1939.21	17315.7
0.8	1.7	0.7	0.011225	0.87109	12.3303	321.659	2014.17
0.8	1.5	0.9	0.002622	0.48225	6.22579	110.804	549.251
0.03	1.6	0.05	0.082033	0.58219	5.44714	112.496	625.448
0.1	1.6	0.2	0.927866	10.3322	103.545	2155.31	11986.9
0.3	1.6	0.4	0.180146	4.04235	44.5251	936.592	5211.2
0.5	1.6	0.7	0.021929	1.66553	21.2695	454.552	2530.79
1	1.6	0.9	0.005511	1.02005	14.4349	311.781	1736.65
0.03	1.6	0.05	0.082033	0.58219	5.44714	112.496	625.448
0.03	1.6	0.2	0.012103	0.13477	1.35059	28.1127	156.351
0.03	1.6	0.4	0.002702	0.06064	0.66788	14.0489	78.168
0.03	1.6	0.7	0.000387	0.02939	0.37534	8.02151	44.661
0.03	1.6	0.9	0.00011	0.0204	0.2887	6.23562	34.733

Table 3. CHRF results for the OEA distribution against selected parametric combinations.

$$\psi_{OEA-G}(t) = \frac{f(t)}{F(t)} \frac{\gamma\theta^2 te^{-2\theta t} \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{\left[1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}\right]^2}}{\exp\left\{\gamma \frac{\left[1-(1+2\theta t)\,e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)\,e^{-2\theta t}\right]^{\lambda}}\right\}-1}. \tag{19}$$

Reviewing the behavior in Fig. 2, it's clear that the CHRF for the OEA distribution generally increases over time. This trend is supported by the numerical results in Table 3, which were derived from Equation (18). The table demonstrates that, overall, the CHRF for the OEA distribution tends to increase as time progresses. However, when the γ value is constant at 0.8, the CHRF decreases as the scale parameter θ decreases (see rows 6 to 10). Conversely,

γ	θ	λ	t				
			0.02	**0.4**	**1.2**	**2.3**	**2.9**
0.03	3	0.01	0.019718	0.0515	0.84934	1.94934	2.54934
0.1	2.1	0.05	0.019999	0.39407	1.04818	1.55932	1.79186
0.3	1.9	0.3	0.019996	0.37544	1.0244	2.11863	2.71863
0.5	1.7	0.6	0.02	0.4	1.2	2.29999	2.89998
1	1.5	0.7	0.02	0.4	1.19999	2.29993	2.89986
0.8	3	0.05	2.02205	768.308	68350.3	167244	5794.34
0.8	2.3	0.2	0.05714	18.6641	726.117	69367.1	162789
0.8	2	0.4	0.00715	2.89416	82.5658	4368.81	38068.6
0.8	1.7	0.7	0.00061	0.53987	11.8663	346.113	2218.97
0.8	1.5	0.9	0.00011	0.20953	4.0337	80.616	410.47
0.03	1.6	0.05	0.01471	3.38246	65.6645	1578.4	8653.45
0.1	1.6	0.2	0.00272	0.70246	13.6796	329.268	1876.07
0.3	1.6	0.4	0.00141	0.52156	10.2585	246.967	1409.92
0.5	1.6	0.7	0.00031	0.27611	5.581	134.406	767.654
1	1.6	0.9	0.00018	0.32528	6.75015	162.615	928.848
0.03	1.6	0.05	0.172093	4.28184	0.28415	2.19714	8.65345
0.03	1.6	0.2	0.066993	1.4607	1.43508	6.10185	5.62822
0.03	1.6	0.4	0.052281	0.83434	1.07974	1.95565	1.58945
0.03	1.6	0.7	0.048636	0.5554	0.73516	0.00265	1.85046
0.03	1.6	0.9	0.0482	0.47346	0.60505	0.00957	2.29459

Table 4. RHRF results for the OEA distribution against selected parametric combinations.

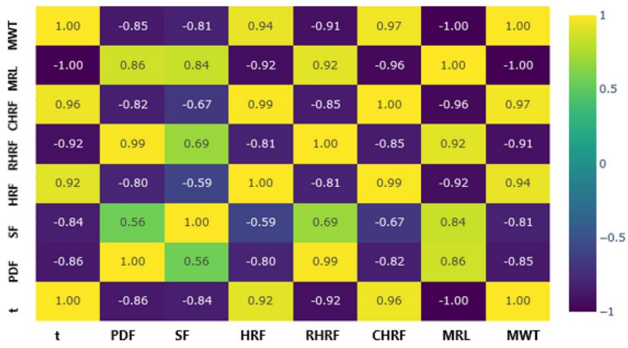
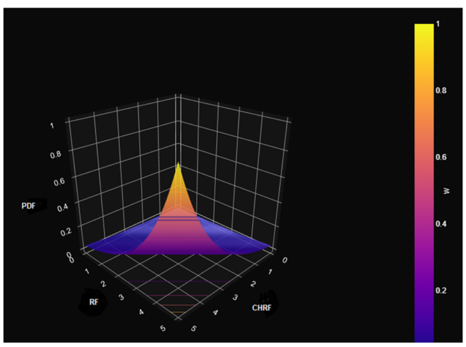


Fig. 3. Heatmap visualization for various reliability measures of the OEA model.

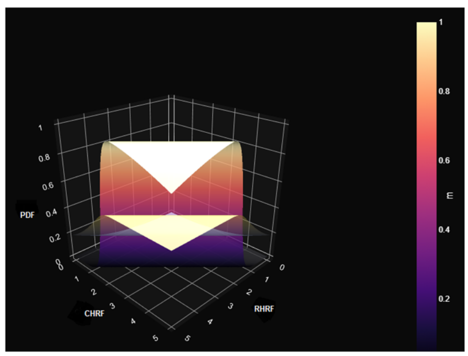
with a fixed θ of 1.6 and small x values (e.g., 0.02, 0.4), the CHRF decreases as γ increases. For large t values, the CHRF seems to become independent of the shape parameter, showing an increasing trend for a fixed θ when $x \geq 1.2$ (see rows 11 to 15). Furthermore, as shown in rows 16 to 20, the CHRF increases when the scale parameter λ is relatively large and decreases when it's smaller, while both γ and θ are fixed at 0.03 and 1.6, respectively.

Likewise, the numerical results from Eqs. (19) are presented in Table 4, confirming that the RHRF for the OEA distribution generally follows an increasing trend over time. For a constant γ value of 0.8, the RHRF decreases as the scale parameter θ decreases (refer to rows 6 to 10 of Table 4). Conversely, with θ fixed at 1.6 and small x values (e.g., 0.02, 0.4), the RHRF decreases with an increase in the γ value. However, for large t values, the RHRF appears to become independent of the shape parameter, showing an increasing trend for a fixed θ when $x \geq 1.2$ (see rows 11 to 15). Additionally, as shown in rows 16 to 20, the RHRF increases when the scale parameter λ is relatively large and decreases when it is smaller, while both γ and θ are fixed at 0.03 and 1.6, respectively.

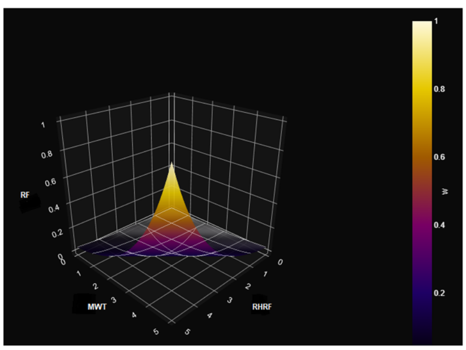
Heat-map in Fig. 3 offers correlation related visualizations for various reliability measures of the proposed OEA distribution. Interesting correlation relationships between various measures are evident from this visual presentation. It indicates how changes in various reliability measures of the OEA model are associated with changes in other measures. Heat-map is widely used in fields like economics, social sciences, medicine, and finance to understand relationships between variables. For example, a company might study the correlation between advertising spending and sales to determine if more advertising leads to higher sales. The coding in Listing 2 offers procedure for obtaining correlation matrix and heat-map related visual presentations of OEA distribution. Besides, Fig. 4



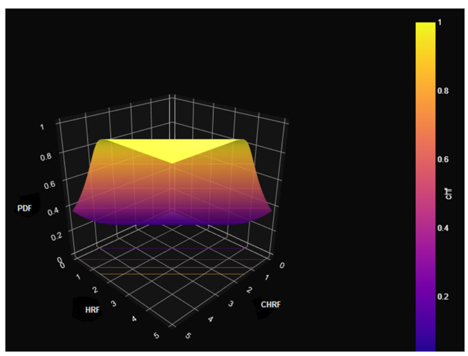
(a) 3D surface plot of PDF, RF, and CHRF



(b) 3D surface plot of PDF, RHRF, and CHRF



(c) 3D surface plot of RF, RHRF, and MWTF



(d) 3D surface plot of PDF, HRF, and CHRF

Fig. 4. 3D surface plot visualizations for various reliability measures of the OEA model.

presents a comprehensive suite of 3D surface plots analyzing the OEA model's reliability measures. The visualizations in Fig. 4a, b, c, and d collectively demonstrate the complex relationships between probability density, hazard functions, and reliability metrics, revealing critical patterns in the model's failure rate behavior and survival characteristics under varying parameter conditions. The coding in Listing 3 offers a procedure for 3D surface plot implementations for different reliability measures of the proposed OEA distribution.

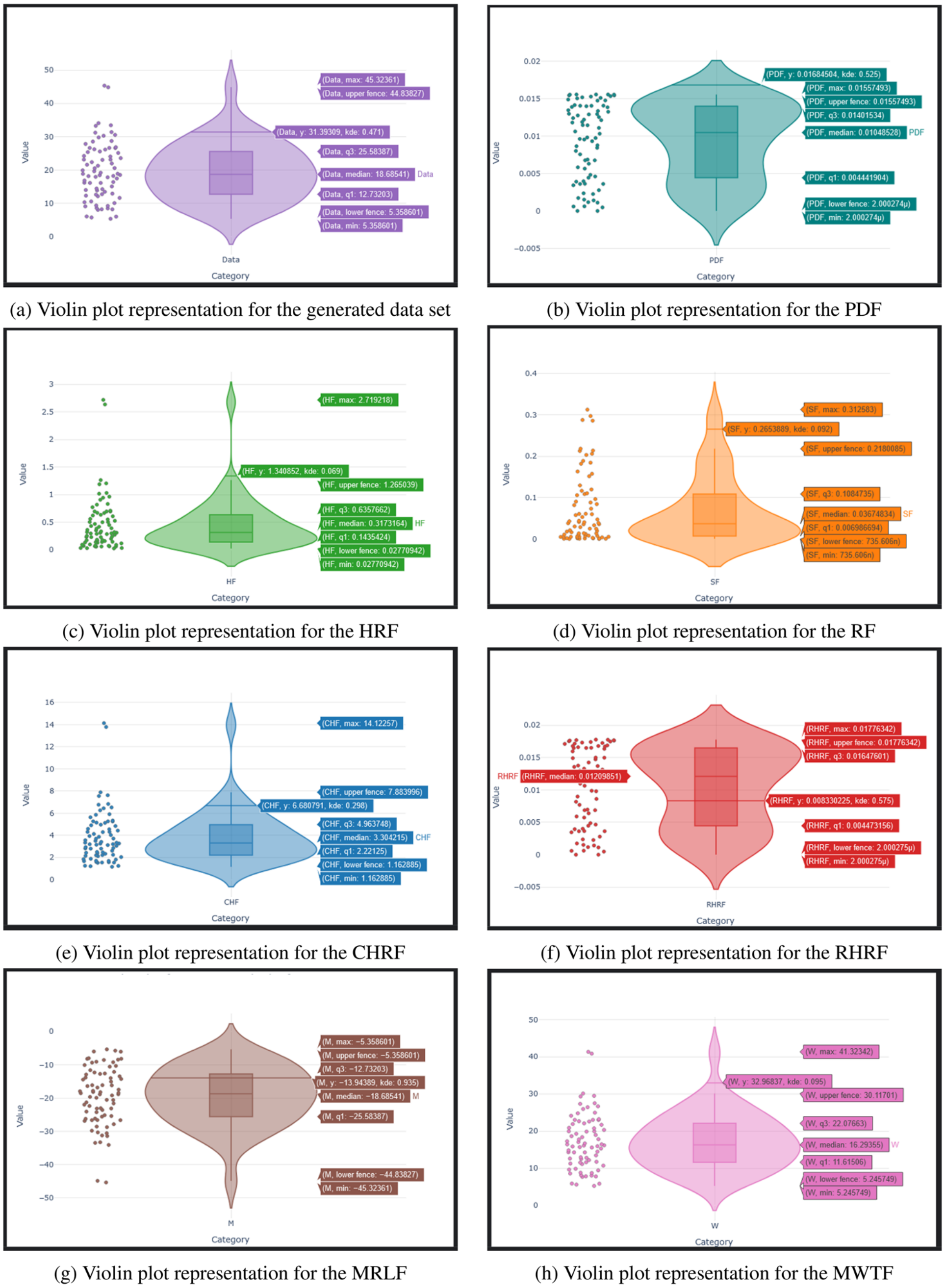
```
def evaluate_functions_on_array(t_array):
pdf_results = []
sf_results = []
hf_results = []
rhrf_results = []
chf_results = []
m_results = []
w_results = []


for x in x_array:
pdf_results.append(beta_pdf(beta_param=0.2, delta=0.03, lambd=0.05, x=x))
sf_results.append(beta_sf(beta_param=0.2, delta=0.03, lambd=0.05, x=x))
hf_results.append(beta_hf(beta_param=0.2, delta=0.03, lambd=0.05, x=x))
rhrf_results.append(beta_rhrf(beta_param=0.2, delta=0.03, lambd=0.05, x=x))
chf_results.append(beta_chf(beta_param=0.2, delta=0.03, lambd=0.05, x=x))
m_results.append(m_function(t=x, beta_param=0.8, delta=1.3, lambd=0.5))
w_results.append(w_function(t=x, beta_param=0.2, delta=0.03, lambd=0.05))

return {
  'pdf': np.array(pdf_results),
  'sf': np.array(sf_results),
  'hf': np.array(hf_results),
  'rhrf': np.array(rhrf_results),
  'chf': np.array(chf_results),
  'm': np.array(m_results),
  'w': np.array(w_results)
}

# Evaluate the functions on the data points
results = evaluate_functions_on_array(data)

# Convert the results to a DataFrame
results_df = pd.DataFrame({
  'x': data,
  'pdf': results['pdf'],
  'sf': results['sf'],
  'hf': results['hf'],
  'rhrf': results['rhrf'],
  'chf': results['chf'],
  'm': results['m'],
  'w': results['w']
})

#print(results_df)

# Compute the correlation matrix
correlation_matrix = results_df.corr()
print(correlation_matrix)

# Create the heatmap
fig = go.Figure(data=go.Heatmap(
z=correlation_matrix.values,
x=correlation_matrix.columns,
y=correlation_matrix.columns,
colorscale='Viridis',
text=correlation_matrix.values,
texttemplate="%{text:.2f}",
textfont={"size":12}
))

# Add title and labels
fig.update_layout(
title='Heatmap of Correlation between Data, PDF, RF, HF, RHRF, CHF, M, and W',
xaxis_title='Variable',
yaxis_title='Variable',
coloraxis_colorbar=dict(title="Correlation")
)

# Show the plot
fig.show()
```

Listing 2: Correlation matrix implementation for OEA distribution

```python
import itertools


# Extract data from the DataFrame
variables = {
  'pdf': results_df['pdf'],
  'sf': results_df['sf'],
  'hf': results_df['hf'],
  'rhrf': results_df['rhrf'],
  'chf': results_df['chf'],
  'm': results_df['m'],
  'w': results_df['w']
}

# Get all possible combinations of the 7 variables taken 3 at a time
combinations = list(itertools.combinations(variables.keys(), 3))

for combo in combinations:
x_var, y_var, z_var = combo
x = variables[x_var]
y = variables[y_var]
z = variables[z_var]

# Create meshgrid for y and z
y, z = np.meshgrid(y, z)

# Create a 3D surface plot
fig = go.Figure(data=[go.Surface(t=x, y=y, z=z)])

# Add title and labels
fig.update_layout(title=f'3D Surface Plot of {x_var}, {y_var}, and {z_var}',
scene=dict(taxis_title=x_var, yaxis_title=y_var, zaxis_title=z_var))

# Show the plot
fig.show()
```

Listing 3: 3D surface plot implementation for OEA distribution

Violin plot representations for various reliability measures of the proposed OEA distribution are shown in Fig. 5, offering a comprehensive visualization of data distribution by integrating features of box plots and density plots. The shape of each violin reflects the PDF, where wider regions indicate higher data concentration and narrower sections suggest lower density. As a symmetrical representation, it mirrors kernel density estimation on both sides of the centerline. While the embedded box plot highlights key summary statistics such as the median, interquartile range (IQR), and potential outliers. This renders violin plots especially handy in the exploration of intricate distributions, such as multimodal forms, where regular box plots will not be able to reflect subtleties in the distributional detail. Comparing more than two groups, changes in shape, median, and spread in violins reflect differences in central tendency, variance, and pattern of distribution across conditions. For example, RF violin plot in Fig. 5d displays reliability data distribution to help with interpreting survival probabilities for various time points. In like manner, violin plot of the HRF in Fig. 5c plots out HR distributions and helps with comparing risk trends at different time intervals. While RHRF violin plot in Fig. 5f presents a reverse view with the use of reverse HR. The CHRF violin plot (Fig. 5e) extends this further by showing the cumulative HR distribution, which captures how the total risk builds up over time. Similarly, the MRLF violin plot (Fig. 5g) emphasizes the distribution of MRL values, providing more give a deeper look intot into the anticipated remaining lifetime under various conditions. Lastly, the MWTF violin plot (Fig. 5h) displays the distribution of MWTF in different parametric groups, which is a



(a) Violin plot representation for the generated data set

(b) Violin plot representation for the PDF

(c) Violin plot representation for the HRF

(d) Violin plot representation for the RF

(e) Violin plot representation for the CHRF

(f) Violin plot representation for the RHRF

(g) Violin plot representation for the MRLF

(h) Violin plot representation for the MWTF

Fig. 5. Violin plot presentations for various reliability measures of the OEA distribution.

crucial parameter in queuing theory as well as in reliability analysis for measuring delays prior to event occurrences. Together, these visualizations provide a detailed statistical representation of reliability measures, facilitating a nuanced interpretation of distributional characteristics across different reliability functions. The coding in Listing 4 offers a procedure for obtaining violin plot-related visual presentations for different measures of the proposed OEA distribution.

```
    df_data = pd.DataFrame({'Value': results_df['x'], 'Category': ['Data'] * len(data
   )})
    df_pdf = pd.DataFrame({'Value': results_df['pdf'], 'Category': ['PDF'] * len(data
   )})
    df_sf = pd.DataFrame({'Value': results_df['sf'], 'Category': ['SF'] * len(data)})
    df_hf = pd.DataFrame({'Value': results_df['hf'], 'Category': ['HF'] * len(data)})
    df_rhrf = pd.DataFrame({'Value': results_df['rhrf'], 'Category': ['RHRF'] * len(
   data)})
    df_chf = pd.DataFrame({'Value': results_df['chf'], 'Category': ['CHF'] * len(data
   )})
    df_m = pd.DataFrame({'Value': results_df['m'], 'Category': ['M'] * len(data)})
    df_w = pd.DataFrame({'Value': results_df['w'], 'Category': ['W'] * len(data)})

    # Define color sequences
    colors = {
      'Data': '#9467bd',  # Purple
      'PDF': '#008080',   # Blue
      'SF': '#ff7f0e',    # Orange
      'HF': '#2ca02c',    # Green
      'RHRF': '#d62728',  # Red
      'CHF': '#1f77b4',   # Light blue
      'M': '#8c564b',     # Brown
      'W': '#e377c2'      # Pink
    }

    # Helper function to create violin plots with additional styling
    def create_violin_plot(df, title, color):
    fig = px.violin(df, y='Value', x='Category', box=True, points="all", title=title,
    color='Category', color_discrete_sequence=[color])

    # Update layout for more attractive styling
    fig.update_layout(
    title=dict(t=0.5, font=dict(size=20)),
    yaxis=dict(title='Value', gridcolor='rgba(0, 0, 0, 0.1)'),
    xaxis=dict(title='Category', gridcolor='rgba(0, 0, 0, 0.1)'),
    plot_bgcolor='white',
    showlegend=False
    )

    # Add shadow effect to violin plot
    fig.update_traces(marker_line_width=0.5, opacity=0.9, selector=dict(type='violin'
   ))

    # Show the plot
    fig.show()

    # Create and show violin plots
    create_violin_plot(df_data, 'Violin Plot for Data', colors['Data'])
    create_violin_plot(df_pdf, 'Violin Plot for PDF', colors['PDF'])
    create_violin_plot(df_sf, 'Violin Plot for RF', colors['SF'])
    create_violin_plot(df_hf, 'Violin Plot for HF', colors['HF'])
    create_violin_plot(df_rhrf, 'Violin Plot for RHRF', colors['RHRF'])
    create_violin_plot(df_chf, 'Violin Plot for CHF', colors['CHF'])
    create_violin_plot(df_m, 'Violin Plot for M', colors['M'])
    create_violin_plot(df_w, 'Violin Plot for W', colors['W'])
```

Listing 4: Violin plot implementations for OEA distribution

Moments, moment-generating function, and characteristic function of the OEA distribution

Moments

The k-th moment of the OEA distribution is defined as:

```
•message: CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
•success: True
•status: 0
•fun: 323.23801868120216
•x: [ 2.705e-01  5.739e-01]
•nit: 10
•jac: [-5.684e-05  1.137e-05]
•nfev: 39
•njev: 13
•hess_inv: <2x2 LbfgsInvHessProduct with dtype=float64>
•MLE: theta=0.27, lambd=0.57
•Inverse Hessian:
•[[0.0003531  0.00112311]
•[0.00112311 0.00516923]]
•VCV(MLE) =
•[[0.0003531  0.00112311]
•[0.00112311 0.00516923]]
•Standard error for theta estimate =  0.01879107987127288
•Standard error for lambda estimate =  0.07189736709206919
•theta_MLE = 0.2704818025294143 , lower bound 95% conf. int. = 0.23289964278686853
•lambda_MLE = 0.5738556707581985 , lower bound 95% conf. int. = 0.4300609365740602
•hypothesis value log likelihood -327.5690645523873
•MLE log likelihood -323.23801868120216
•likelihood ratio value 8.662091742370308
•chi squared of H0 with 2 degrees of freedom p-value =  0.013153783130708807
```

Fig. 6. Optimization Results for the OEA MLE.

$$\begin{aligned}\mu_k =&\mathbb{E}[T^k] = \int_{-\infty}^{\infty} t^k f_{\text{OEA-G}}(t)\, dt,\\ =&\int_{-\infty}^{\infty} t^k \gamma g(t,\theta) \frac{G^{\lambda-1}(t,\theta)}{[1-G^{\lambda}(t,\theta)]^2} \exp\left\{-\gamma \frac{G^{\lambda}(t,\theta)}{1-G^{\lambda}(t,\theta)}\right\} dt,\end{aligned} \tag{20}$$

where $f_{\text{OEA-G}}(t)$ is the PDF of the OEA distribution, given by:

$$f_{\text{OEA-G}}(t) = \gamma g(t,\theta) \frac{G^{\lambda-1}(t,\theta)}{[1-G^{\lambda}(t,\theta)]^2} \exp\left\{-\gamma \frac{G^{\lambda}(t,\theta)}{1-G^{\lambda}(t,\theta)}\right\},$$

with parameters $\gamma > 0, \theta > 0$, and $\lambda > 0$, and where $g(t,\theta)$ is the PDF of the baseline distribution, and $G(t,\theta)$ is its CDF. For the specific case where $G(t,\theta) = 1-(1+2\theta t)e^{-2\theta t}$, the moment becomes:

$$\mu_k = \int_0^{\infty} t^k \gamma \theta^2 t e^{-2\theta t} \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{\left[1-[1-(1+2\theta t)e^{-2\theta t}]^{\lambda}\right]^2} \exp\left\{-\gamma \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1-[1-(1+2\theta t)e^{-2\theta t}]^{\lambda}}\right\} dt.$$

This integral generally lacks a closed form due to the complexity of the integrand, particularly the fractional and exponential terms. Numerical methods, such as Monte Carlo integration or quadrature, are recommended for evaluation. For small λ, the distribution may approximate the baseline distribution's moments, adjusted by the skewness introduced by λ.

The mean ($k=1$) is:

$$\begin{aligned}\mu_1 =&\mathbb{E}[T] = \int_0^{\infty} t f_{\text{OEA-G}}(t)\, dt,\\ =&\int_0^{\infty} t \gamma \theta^2 t e^{-2\theta t} \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{\left[1-[1-(1+2\theta t)e^{-2\theta t}]^{\lambda}\right]^2} \exp\left\{-\gamma \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1-[1-(1+2\theta t)e^{-2\theta t}]^{\lambda}}\right\} dt.\end{aligned} \tag{21}$$

This integral is typically evaluated numerically due to the non-linear combination of exponential and rational functions. The mean is influenced by γ, which controls the scale, and λ, which introduces skewness, shifting the mean relative to the baseline distribution.

The corresponding variance of the proposed distribution is given by:

$$\text{Var}(T) = \mu_2 - \mu_1^2, \tag{22}$$

where the second moment $\mu_2 = \mathbb{E}[T^2]$ is:

$$\mu_2 = \int_0^\infty t^2\gamma\theta^2 te^{-2\theta t}\frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^\lambda}{\left[1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^\lambda\right]^2}\exp\left\{-\gamma\frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^\lambda}{1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^\lambda}\right\}dt.$$

Similar to the mean, numerical integration is required. The variance reflects the spread of the distribution, which is modulated by γ, θ, and λ.

Skewness (γ_1) and kurtosis (γ_2) of the proposed model are respectively computed as:

$$\gamma_1 = \frac{\mu_3 - 3\mu_1\mu_2 + 2\mu_1^3}{(\mu_2-\mu_1^2)^{3/2}}, \quad \gamma_2 = \frac{\mu_4 - 4\mu_1\mu_3 + 6\mu_1^2\mu_2 - 3\mu_1^4}{(\mu_2-\mu_1^2)^2},$$

where μ_3 and μ_4 are the third and fourth moments, computed similarly as:

$$\mu_3 = \int_0^\infty t^3 f_{\text{OEA-G}}(t)\,dt, \quad \mu_4 = \int_0^\infty t^4 f_{\text{OEA-G}}(t)\,dt.$$

These higher moments capture the asymmetry and tail behavior of the OEA distribution. For $\lambda > 1$, positive skewness is expected, making the distribution suitable for modeling right-skewed data, such as failure times or financial returns. High kurtosis indicates heavier tails, useful in applications like risk modeling. To address the complexity of the integral forms and provide explicit mathematical derivations, we derive series expansions for the key moments of the OEA distribution. Let $T \sim \text{OEA-G}(\gamma,\lambda,\theta)$, with baseline CDF $G(t,\theta) = 1-(1+2\theta t)e^{-2\theta t}$ and PDF $g(t,\theta) = 4\theta^2 te^{-2\theta t}$, $t \geq 0$. The PDF of the OEA distribution is:

$$f_{\text{OEA}}(t) = \gamma g(t,\theta)\frac{G^{\lambda-1}(t,\theta)}{[1-G^\lambda(t,\theta)]^2}\exp\left\{-\gamma\frac{G^\lambda(t,\theta)}{1-G^\lambda(t,\theta)}\right\}.$$

Define the odds ratio transformation:

$$u = \frac{G(t,\theta)}{1-G(t,\theta)} \quad \Rightarrow \quad G(t,\theta) = \frac{u}{1+u}, \quad 1-G(t,\theta) = \frac{1}{1+u}.$$

Then the exponential term becomes:

$$\exp\left\{-\gamma\frac{G^\lambda}{1-G^\lambda}\right\} = \exp\left(-\gamma u^\lambda\right).$$

The PDF can be re-expressed in terms of u, but we instead exploit the series structure of the OEA generator.

The OEA density involves the Generator term:

$$\frac{G^{\lambda-1}(t,\theta)}{[1-G^\lambda(t,\theta)]^2}\exp\left\{-\gamma\frac{G^\lambda(t,\theta)}{1-G^\lambda(t,\theta)}\right\}.$$

Let $w = G(t,\theta)$, so $0 < w < 1$. Then:

$$f_{\text{OEA}}(t) = \gamma g(t,\theta)\cdot\frac{w^{\lambda-1}}{(1-w^\lambda)^2}e^{-\gamma\frac{w^\lambda}{1-w^\lambda}},$$

now expand:

$$\frac{1}{(1-w^\lambda)^2} = \sum_{m=1}^\infty m w^{\lambda(m-1)}, \quad |w^\lambda| < 1.$$

This is the generating function for $\sum m x^{m-1} = \frac{1}{(1-x)^2}$.

Thus:

$$\frac{w^{\lambda-1}}{(1-w^\lambda)^2} = w^{\lambda-1}\sum_{m=1}^\infty m w^{\lambda(m-1)} = \sum_{m=1}^\infty m w^{\lambda m-1}.$$

Also expand the exponential:

$$\exp\left(-\gamma\frac{w^\lambda}{1-w^\lambda}\right) = \exp\left(-\gamma\sum_{j=1}^\infty w^{\lambda j}\right) = \prod_{j=1}^\infty \exp(-\gamma w^{\lambda j}).$$

Using $\exp(-ax) = \sum_{k=0}^\infty \frac{(-ax)^k}{k!}$, we get:

$$\exp\left(-\gamma\frac{w^{\lambda}}{1-w^{\lambda}}\right)=\sum_{n=0}^{\infty}\frac{(-\gamma)^n}{n!}\left(\sum_{j=1}^{\infty}w^{\lambda j}\right)^n.$$

Using the multinomial theorem, this becomes a multiple sum over compositions of n into parts of size λj.
Combining above expansions we obtain the series representation of the PDF as:

$$f_{\mathrm{OEA}}(t)=\gamma g(t,\theta)\sum_{m=1}^{\infty}mw^{\lambda m-1}\sum_{n=0}^{\infty}\frac{(-\gamma)^n}{n!}\left(\sum_{j=1}^{\infty}w^{\lambda j}\right)^n.$$

This is complex, so we truncate and recombine using a unified series approach by defining the OEA weight function as:

$$\phi(w)=\frac{w^{\lambda-1}}{(1-w^{\lambda})^2}\exp\left(-\gamma\frac{w^{\lambda}}{1-w^{\lambda}}\right).$$

We expand $\phi(w)$ as:

$$\phi(w)=\sum_{p=0}^{\infty}c_p w^p,$$

where coefficients c_p can be computed recursively or numerically. Hence, the closed-form series for the k-th raw moment is:

$$\mu_k=\mathbb{E}[T^k]=\int_0^{\infty}t^k f_{\mathrm{OEA}}(t)\,dt=\gamma\int_0^{\infty}t^k g(t,\theta)\phi(G(t,\theta))\,dt.$$

Make the substitution $w=G(t,\theta)$, so $t=G^{-1}(w,\theta)$ and $dt=\frac{dw}{g(G^{-1}(w,\theta),\theta)}$. Let $h(w)=G^{-1}(w,\theta)$, so k^{th} raw moment is computed as:

$$\mu_k=\gamma\int_0^1[h(w)]^k\phi(w)\,dw.$$

Now insert the series:

$$\phi(w)=\sum_{p=0}^{\infty}c_p w^p\quad\Rightarrow\quad\mu_k=\gamma\sum_{p=0}^{\infty}c_p\int_0^1[h(w)]^k w^p\,dw=\gamma\sum_{p=0}^{\infty}c_p\mathbb{E}_w[h(w)^k w^p],$$

where the expectation is over $w\sim\mathrm{Beta}(p+1,1)$ if approximated, but exactly:

$$\int_0^1 h(w)^k w^p\,dw.$$

To obtain the explicit series for moments using baseline QF the baseline inverse CDF $h(w)=G^{-1}(w,\theta)$, solve:

$$w=1-(1+2\theta t)e^{-2\theta t}\quad\Rightarrow\quad(1+2\theta t)e^{-2\theta t}=1-w.$$

Let $z=2\theta t$, so the above expression is written as:

$$(z+1)e^{-z}=1-w.$$

This is solved using the Lambert W function:

$$z=-W(-(1-w)e^{-1})-1\quad\Rightarrow\quad t=\frac{-W(-(1-w)e^{-1})-1}{2\theta}.$$

Thus:

$$h(w)=\frac{-W(-(1-w)e^{-1})-1}{2\theta},$$

now expand $W(x)$ for small x:

$$W(x)=x-x^2+\frac{3}{2}x^3-\frac{8}{3}x^4+\cdots,\quad |x|<\frac{1}{e}.$$

Let $x = -(1-w)e^{-1}$, so:

$$W(x) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} x^n.$$

Thus:

$$h(w) = \frac{-1}{2\theta} \left[1 + \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} \left(-(1-w)e^{-1} \right)^n \right].$$

This allows term-by-term integration in the moment integral. Series expansion of μ_k is:

$$\mu_k = \gamma \sum_{p=0}^{\infty} c_p \int_0^1 h(w)^k w^p \, dw.$$

Using $h(w) = \sum_{q=0}^{\infty} d_q (1-w)^q$, we get:

$$\mu_k = \gamma \sum_{p=0}^{\infty} \sum_{r=0}^{\infty} c_p d_r^{(k)} \int_0^1 (1-w)^r w^p \, dw = \gamma \sum_{p=0}^{\infty} \sum_{r=0}^{\infty} c_p d_r^{(k)} B(p+1, r+1),$$

where $B(p,r) = \frac{\Gamma(p)\Gamma(r)}{\Gamma(p+r)}$.

Thus:

$$\mu_k = \gamma \sum_{p=0}^{\infty} \sum_{r=0}^{\infty} c_p d_r^{(k)} \frac{\Gamma(p+1)\Gamma(r+1)}{\Gamma(p+r+2)}.$$

This is a double series in computable coefficients. This expresses the k-th moment of the OEA distribution as an infinite weighted sum of the moments of the AD, which are more straightforward to compute either analytically or numerically. The same approach can be applied to derive other properties, such as the moment generating function, incomplete moments, and mean residual life, transforming them from intractable integral forms into computationally feasible series representations. For practical computation, the infinite series can be truncated after a sufficient number of terms to achieve the desired precision. This series expansion methodology significantly enhances the mathematical tractability of the OEA distribution and facilitates its application in real-world data analysis.

Moment-generating function

The moment-generating function (MGF) is defined as:

$$M(s) = \mathbb{E}[e^{sT}] = \int_0^{\infty} e^{st} f_{\text{OEA-G}}(t) \, dt, \tag{23}$$

or explicitly:

$$M(s) = \int_0^{\infty} e^{st} \gamma \theta^2 t e^{-2\theta t} \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{\left[1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}\right]^2} \exp\left\{ -\gamma \frac{\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1-\left[1-(1+2\theta t)e^{-2\theta t}\right]^{\lambda}} \right\} dt.$$

Due to the complex structure of the integrand, a closed-form expression is generally not feasible. For small s, the MGF can be approximated numerically or via series expansion:

$$M(s) \approx \sum_{k=0}^{n} \frac{s^k}{k!} \mu_k,$$

where μ_k are the moments computed above. Hence we get:

$$M(s) = \mathbb{E}[e^{sT}] = \gamma \int_0^1 e^{sh(w)} \phi(w) \, dw = \gamma \sum_{p=0}^{\infty} c_p \int_0^1 e^{sh(w)} w^p \, dw.$$

Expand $e^{sh(w)} = \sum_{m=0}^{\infty} \frac{(sh(w))^m}{m!}$, so above expression can be written as:

Parameter	Value
Convergence Message	CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
Success	True
Status	0
Objective Function (fun)	323.238019
Initial Parameters (x)	[0.2705, 0.5739]
Iterations (nit)	10
Jacobian (jac)	[-5.684e-05, 1.137e-05]
Function Evaluations (nfev)	39
Jacobian Evaluations (njev)	13
MLE	$\theta = 0.27, \lambda = 0.57$
Inverse Hessian	[[0.0003531, 0.00112311], [0.00112311, 0.00516923]]
VCV(MLE)	[[0.0003531, 0.00112311], [0.00112311, 0.00516923]]
Standard Error (θ)	0.018791
Standard Error (λ)	0.071897
θ_{MLE}	0.270482
λ_{MLE}	0.573856
95% CI Lower Bound (θ)	0.232900
95% CI Lower Bound (λ)	0.430061
Hypothesis Log-Likelihood	-327.569065
MLE Log-Likelihood	-323.238019
Likelihood Ratio	8.662092
p-value (Chi-squared, df=2)	0.013154

Table 5. Optimization Results for the MLE of OEA distribution.

$$M(s) = \gamma \sum_{p=0}^{\infty} \sum_{m=0}^{\infty} \frac{s^m}{m!} c_p \int_0^1 h(w)^m w^p \, dw = \sum_{m=0}^{\infty} \frac{s^m}{m!} \mu_m.$$

Alternatively, MGF can also be computed as:

$$M(s) = \gamma \sum_{p=0}^{\infty} c_p \sum_{m=0}^{\infty} \frac{s^m}{m!} \mathbb{E}[h(w)^m | w \sim \text{Beta}(p+1, 1)].$$

The MGF is valuable in applications like reliability analysis, where it aids in characterizing the distribution's behavior under transformations.

Characteristic function
The characteristic function is defined as $\psi(s) = \mathbb{E}[e^{isT}]$, and for the OEA distribution it is computed as:

$$\psi(s) = \int_0^{\infty} e^{ist} f_{\text{OEA-G}}(t) \, dt, \tag{24}$$

or

$$\psi(s) = \int_0^{\infty} e^{ist} \gamma \theta^2 t e^{-2\theta t} \frac{\left[1 - (1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{\left[1 - \left[1 - (1+2\theta t)e^{-2\theta t}\right]^{\lambda}\right]^2} \exp\left\{ -\gamma \frac{\left[1 - (1+2\theta t)e^{-2\theta t}\right]^{\lambda}}{1 - \left[1 - (1+2\theta t)e^{-2\theta t}\right]^{\lambda}} \right\} dt.$$

Like the MGF, the characteristic function lacks a closed form due to the interplay of exponential and rational terms. It can be approximated numerically or through the moments:

$$\psi(s) \approx \sum_{k=0}^{n} \frac{(is)^k}{k!} \mu_k.$$

Characteristic function may be rewritten as:

$$\psi(s) = \mathbb{E}[e^{isT}] = \gamma \sum_{p=0}^{\infty} c_p \int_0^1 e^{ish(w)} w^p \, dw.$$

Category	Parameter/Function	Value
Statistical Moments	Mean	0.2731
	Variance	0.6023
	Skewness	2.8267
	Kurtosis	9.9340
Moment-Generating Function	$M(s=-0.10)$	0.1008
	$M(s=-0.05)$	0.1122
	$M(s=0.00)$	0.1250
	$M(s=0.05)$	0.1395
	$M(s=0.10)$	0.1560
Characteristic Function	$\phi(s=-0.10)$	$0.1216-0.0270i$
	$\phi(s=-0.05)$	$0.1242-0.0136i$
	$\phi(s=0.00)$	$0.1250+0.0000i$
	$\phi(s=0.05)$	$0.1242+0.0136i$
	$\phi(s=0.10)$	$0.1216+0.0270i$

Table 6. Comprehensive Statistical Characterization of the OEA Distribution Parameters: $\gamma=1.0$, $\theta=0.5$, $\lambda=2.0$.

Using $e^{ish(w)}=\cos(sh(w))+i\sin(sh(w))$, and expanding:

$$\psi(s)=\gamma\sum_{p=0}^{\infty}c_p\left[\sum_{m=0}^{\infty}\frac{(-1)^m(sh(w))^{2m}}{(2m)!}+i\sum_{m=0}^{\infty}\frac{(-1)^m(sh(w))^{2m+1}}{(2m+1)!}\right]\text{(integrated term-by-term)}.$$

In applications such as signal processing or financial modeling, the characteristic function is useful for analyzing distributional convergence in Fourier space, particularly for assessing tail behavior and symmetry.

The moments, MGF, and characteristic function of the OEA distribution provide give a deeper look intots into its statistical properties. While analytical solutions are limited by the complexity of the integrands, numerical methods enable practical computation. These properties make the OEA distribution suitable for modeling skewed and heavy-tailed data in fields like reliability engineering and financial risk analysis. The numerical results are summarized in Tables 6 which details the statistical properties, MGF values, and characteristic function values, respectively.

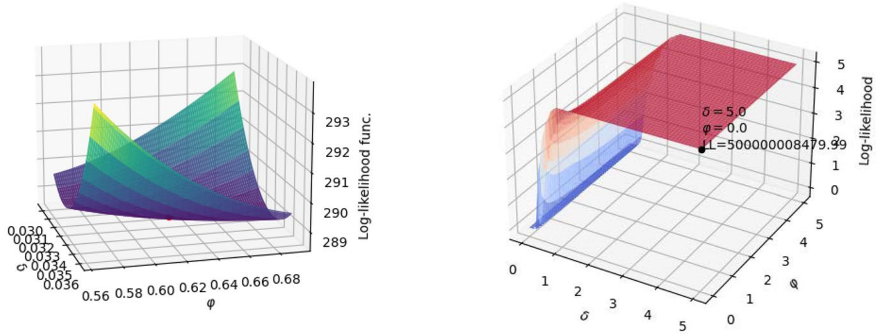
The statistical properties presented in Table 6 reveal key characteristics of the OEA distribution. The mean value of 0.2731 indicates the central tendency of the distribution, while the variance of 0.6023 suggests moderate spread. The skewness of 2.8267 indicates a significant right skew, implying that the distribution has a longer tail on the positive side, which is visually confirmed by the OEA distribution's flexibility exhibited in Fig. 1. The kurtosis of 9.9340, well above 3, suggests a leptokurtic distribution with heavy tails, indicating a higher probability of extreme values compared to a normal distribution. The MGF values in Table 6 show a consistent increase from 0.1008 at $s=-0.10$ to 0.1560 at $s=0.10$, reflecting the distribution's behavior under exponential transformation. This monotonic increase is expected and aligns with the theoretical properties of the OEA distribution, providing a basis for further moment-based analysis. The characteristic function (CF) values in Table 6 exhibit a symmetric pattern around $s=0.00$, where the imaginary component is zero (0.1250 + 0.0000i), confirming the real-valued nature of the distribution at the origin. The imaginary components increase symmetrically with $|s|$, reaching $\pm0.0270i$ at $s=\pm0.10$, which is consistent with the oscillatory behavior of the CF and supports the distribution's complex Fourier representation. These results provide a robust foundation for modeling applications in the context of the research. The analysis demonstrates that the OEA distribution, with the specified parameters, exhibits significant skewness and kurtosis, making it suitable for modeling data with heavy tails and asymmetric behavior. The MGF and CF values further validate its mathematical properties, while the graphical outputs offer intuitive give a deeper look intots.

Kullback-Leibler divergence and variational lower bound

Kullback-Leibler divergence

The Kullback-Leibler (KL) divergence is a fundamental metric in statistical inference and machine learning, particularly in variational autoencoders (VAEs) and generative adversarial networks (GANs), for quantifying the discrepancy between the OEA distribution ($f_{\text{OEA-G}}$) and a reference distribution ($q(t)$). The KL divergence measures the information loss when approximating the OEA distribution with a simpler distribution, such as a Gamma distribution $q(t)=\frac{\beta^\alpha}{\Gamma(\alpha)}t^{\alpha-1}e^{-\beta t}$ for $t>0$, which is suitable given the positive support of the OEA distribution. This is essential for variational inference in reliability modeling and survival analysis. The KL Divergence is defined as:

$$\mathrm{KL}(q||f_{\text{OEA-G}})=\int_0^\infty q(t)\log\left(\frac{q(t)}{f_{\text{OEA-G}}(t)}\right)dt. \tag{25}$$



(a) Log-likelihood plot for selected parameters (b) 3D surface plot of the OEA likelihood

Fig. 7. Visual presentations for the likelihood function of OEA distribution.

Substituting the PDFs, where $q(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}$, we get:

$$\mathrm{KL}(q||f_{\text{OEA-G}}) = \int_0^\infty q(t) \left[\log\left(\frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t} \right) - \log\left(\gamma g(t,\theta) \frac{G^{\lambda-1}(t,\theta)}{[1-G^\lambda(t,\theta)]^2} \exp\left\{ -\gamma \frac{G^\lambda(t,\theta)}{1-G^\lambda(t,\theta)} \right\} \right) \right] dt. \tag{26}$$

Expanding the integrand:

$$\log\left(\frac{q(t)}{f_{\text{OEA-G}}(t)} \right) = \log\left(\frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t} \right) - \log\left(\gamma \theta^2 t e^{-2\theta t} \frac{\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda}{\left[1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda\right]^2} \exp\left\{ -\gamma \frac{\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda}{1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda} \right\} \right).$$

This simplifies to:

$$\begin{aligned} \log\left(\frac{q(t)}{f_{\text{OEA-G}}(t)} \right) =& \alpha \log\beta - \log\Gamma(\alpha) + (\alpha-1)\log t - \beta t \\ & - \log\gamma - \log(\theta^2 t) + 2\theta t + \gamma \frac{\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda}{1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda} \\ & - \lambda \log\left[1-(1+2\theta t)\, e^{-2\theta t}\right] + 2\log\left\{1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda\right\}. \end{aligned}$$

Thus, the KL Divergence is:

$$\begin{aligned} \mathrm{KL}(q||f_{\text{OEA-G}}) = \int_0^\infty q(t) \Big[& \alpha\log\beta - \log\Gamma(\alpha) + (\alpha-1)\log t - \beta t - \log\gamma - \log(\theta^2 t) + 2\theta t \\ & + \gamma \frac{\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda}{1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda} - \lambda\log\left[1-(1+2\theta t)\, e^{-2\theta t}\right] \\ & + 2\log\left\{1-\left[1-(1+2\theta t)\, e^{-2\theta t}\right]^\lambda\right\}\Big]\, dt. \end{aligned} \tag{27}$$

Due to the complexity of the fractional and exponential terms, this integral lacks a closed-form solution. In practice, it is approximated using Monte Carlo methods:

$$\mathrm{KL}(q||f_{\text{OEA-G}}) \approx \frac{1}{n} \sum_{i=1}^n \log\left(\frac{q(t_i)}{f_{\text{OEA-G}}(t_i)} \right), \quad t_i \sim q(t). \tag{28}$$

This approximation is efficient for VAEs and other ML tasks, leveraging samples from $q(t)$. This divergence guides the optimization in variational inference, enhancing the approximation quality in survival models.

For two OEA distributions with parameters $(\gamma_1, \theta_1, \lambda_1)$ and $(\gamma_2, \theta_2, \lambda_2)$, the KL divergence is:

$$D_{\mathrm{KL}}(f_1 \parallel f_2) = \int_0^\infty f_1(t) \ln \frac{f_1(t)}{f_2(t)}\, dt. \tag{29}$$

Substituting the PDFs:

$$\begin{aligned} f_1(t) =& \gamma_1 \theta_1^2 t e^{-2\theta_1 t} \frac{\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}}{\left[1-\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}\right]^2} \exp\left\{-\gamma_1 \frac{\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}}{1-\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}}\right\}, \\ f_2(t) =& \gamma_2 \theta_2^2 t e^{-2\theta_2 t} \frac{\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}{\left[1-\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}\right]^2} \exp\left\{-\gamma_2 \frac{\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}{1-\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}\right\}, \end{aligned}$$

The log-ratio is:

$$\begin{aligned} \ln \frac{f_1(t)}{f_2(t)} =& \ln \frac{\gamma_1}{\gamma_2} + \ln \frac{\theta_1^2 t e^{-2\theta_1 t}}{\theta_2^2 t e^{-2\theta_2 t}} + \ln \frac{\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}}{\left[1-\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}\right]^2} - \ln \frac{\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}{\left[1-\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}\right]^2} \\ &- \gamma_1 \frac{\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}}{1-\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}} + \gamma_2 \frac{\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}{1-\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}. \end{aligned}$$

Thus:

$$\begin{aligned} D_{\mathrm{KL}}(f_1 \parallel f_2) =& \mathbb{E}_{f_1}\left[\ln \frac{\gamma_1}{\gamma_2} + \ln \frac{\theta_1^2}{\theta_2^2} - 2\theta_1 t + 2\theta_2 t - \gamma_1 \frac{\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}}{1-\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}}\right. \\ &+\lambda_1 \log\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right] - 2\log\left[1-\left[1-(1+2\theta_1 t)\, e^{-2\theta_1 t}\right]^{\lambda_1}\right] \\ &-\lambda_2 \log\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right] + 2\log\left[1-\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}\right] \\ &\left.+\gamma_2 \frac{\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}{1-\left[1-(1+2\theta_2 t)\, e^{-2\theta_2 t}\right]^{\lambda_2}}\right]. \end{aligned} \tag{30}$$

This integral is computed numerically due to the complex fractional and exponential terms. In survival analysis, minimizing KL divergence helps in model selection and parameter estimation, improving the fit to observed data.

Evidence lower bound

The Evidence Lower Bound (ELBO) in variational inference for the OEA distribution is given by:

$$\mathrm{ELBO} = \mathbb{E}_q[\log p(T, z)] - \mathrm{KL}(q(z|T)||p(z)), \tag{31}$$

where $p(T, z)$ is the joint distribution, $q(z|T)$ is the variational approximation, and $p(z)$ is the prior. Maximizing the ELBO approximates the posterior in Bayesian models involving the OEA distribution.

We examine the KL divergence between the OEA distribution and the Gamma distribution, as well as between two distinct OEA distributions. The analysis leverages numerical integration and Monte Carlo approximation

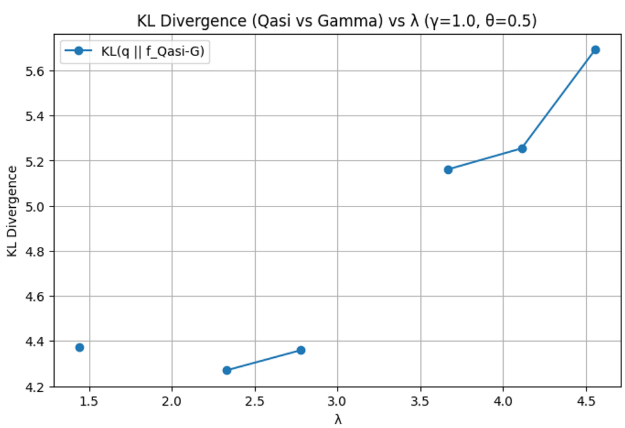


Fig. 8. KL divergence between OEA and Gamma distributions as a function of λ, with $\gamma = 1.0$ and $\theta = 0.5$. The plot highlights the divergence trend across varying λ values.

Comparison	KL divergence Value
Gamma || OEA	4.2830
OEA1 || OEA2	1.4000

Table 7. KL divergence values comparing the Gamma distribution ($\alpha = 2.0, \beta = 1.0$) with the OEA distribution ($\gamma = 1.0, \theta = 0.5, \lambda = 2.0$), and between two OEA distributions (OEA1: $\gamma = 1.0, \theta = 0.5, \lambda = 2.0$; OEA2: $\gamma = 1.5, \theta = 0.7, \lambda = 3.0$).

techniques, with results presented both numerically and graphically. The parameters used are $\gamma = 1.0, \theta = 0.5, \lambda = 2.0$ for the first OEA distribution, and $\gamma = 1.5, \theta = 0.7, \lambda = 3.0$ for the second OEA distribution, with Gamma distribution parameters $\alpha = 2.0$ and $\beta = 1.0$. The graphical results provide visual give a deeper look intots into the KL divergence and distribution comparisons. Figure 8 illustrates the KL divergence between the OEA and Gamma distributions as a function of λ.

The numerical results of the KL divergence analysis are summarized in Table 7, detailing the divergence between the Gamma and OEA distributions, as well as between two OEA distributions.

The KL divergence values presented in Table 7 quantify the dissimilarity between the distributions under study. The KL divergence of 4.2830 between the Gamma distribution and the OEA distribution indicates a substantial difference in their probabilistic structures, suggesting that the OEA distribution with parameters $\gamma = 1.0$, $\theta = 0.5$, and $\lambda = 2.0$ deviates significantly from the Gamma distribution ($\alpha = 2.0$, $\beta = 1.0$). The KL divergence of 1.4000 between the two OEA distributions (OEA1 and OEA2) reflects a moderate difference, attributable to variations in the parameters γ, θ, and λ. This suggests that while the two OEA distributions share a similar functional form, the parameter shifts result in noticeable probabilistic differences. The trend of KL divergence with respect to λ, as depicted in Fig. 8, further indicates that the divergence increases with λ, highlighting the sensitivity of the OEA distribution to this parameter. The runtime warnings observed during computation (e.g., divide by zero and overflow) are indicative of numerical instability in the Monte Carlo approximation, particularly when the denominator $1 - G^{\lambda}$ approaches zero or when the PDF values become excessively large. These issues, as noted in the code output, suggest the need for robust numerical handling, such as adaptive integration bounds or regularization techniques, to ensure reliable estimates in future analysis.

Shannon entropy

The Shannon entropy quantifies the average uncertainty or information content. For the OEA distribution, it is defined as:

$$H(f_{OEA}) = -\int_0^{\infty} f_{OEA-G}(t) \log f_{OEA-G}(t) dt. \tag{32}$$

Expanding the log-PDF, the entropy can be expressed as:

$$\begin{aligned} H(f_{OEA}) = &-\log\gamma - \mathbb{E}[\log g(T,\theta)] - (\lambda-1)\mathbb{E}[\log G(T,\theta)] \\ &+ 2\mathbb{E}\left[\log\left(1 - G^{\lambda}(T,\theta)\right)\right] + \gamma\mathbb{E}\left[\frac{G^{\lambda}(T,\theta)}{1 - G^{\lambda}(T,\theta)}\right], \end{aligned} \tag{33}$$

where expectations are with respect to $T \sim f_{OEA-G}(t)$. A practical method for estimation is Monte Carlo simulation:

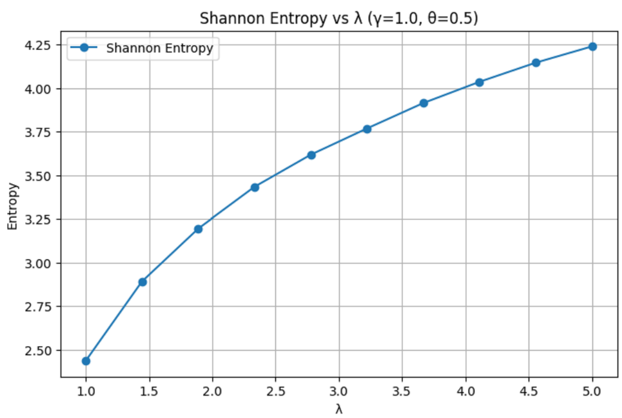


Fig. 9. Shannon entropy of the OEA distribution as a function of λ, with $\gamma = 1.0$ and $\theta = 0.5$. The plot demonstrates the entropy's dependence on the λ parameter.

Parameter Set	Shannon Entropy
$\gamma = 1.0, \theta = 0.5, \lambda = 2.0$	3.2716

Table 8. Shannon entropy value for the OEA distribution with parameters $\gamma = 1.0$, $\theta = 0.5$, and $\lambda = 2.0$, computed using Monte Carlo approximation.

$$H(f_{OEA}) \approx -\frac{1}{n}\sum_{i=1}^{n} \log f_{OEA-G}(t_i), \quad \text{where } t_i \sim f_{OEA-G}(t). \tag{34}$$

We present an analysis of the Shannon entropy for the OEA distribution, computed using Monte Carlo approximation techniques based on numerical integration. The study explores the entropy's behavior with respect to the parameter λ, alongside graphical representations of entropy trends and PDF comparisons. The baseline parameters used are $\gamma = 1.0$, $\theta = 0.5$, and $\lambda = 2.0$, with additional variations in λ to assess sensitivity. The visual results provide give a deeper look intots into the Shannon entropy and the OEA distribution's density profiles. Figure 9 illustrates the variation of Shannon entropy as a function of λ. The numerical findings of the Shannon entropy analysis are summarized in Table 8, detailing the entropy value for the specified parameter set.

The Shannon entropy value of 3.2716, as reported in Table 8, quantifies the uncertainty or information content of the OEA distribution with parameters $\gamma = 1.0$, $\theta = 0.5$, and $\lambda = 2.0$. This moderate entropy value suggests a balanced distribution with neither excessive randomness nor overly constrained variability. The runtime warning regarding a divide by zero encountered in the computation of $W = G^{\lambda}/(1 - G^{\lambda})$ indicates potential numerical instability when G^{λ} approaches 1, necessitating careful handling of the integration bounds or regularization in future iterations. The trend of Shannon entropy with respect to λ, as depicted in Fig. 9, reveals its sensitivity to this parameter. As λ increases, the entropy appears to exhibit a non-linear behavior, potentially reflecting changes in the distribution's tail heaviness or concentration. The Monte Carlo approximation method, while effective, is subject to sampling variability and numerical issues, as evidenced by the warning. To enhance reliability, future analysis could incorporate adaptive sampling techniques or increase the number of samples (n) to mitigate such instabilities. Nonetheless, the current results provide a robust initial assessment of the OEA distribution's entropy characteristics.

Inference of the maximum likelihood estimation

The MLE is the most commonly used method for parameter estimation of distributions. So, we employ MLE for the estimation of unknown parameters of OEA distribution. Let θ denote the p dimensional baseline parametric vector in Equation (4). Let the independent random variables $X_1, X_2 ..., X_n$, come from OEA distribution having the parametric vector $\Theta = (\gamma, \theta, \lambda)^{'}$. Then the likelihood $L(\theta)$ and the log-likelihood $l(\theta)$ is obtained from Equation (6) as

$$\begin{aligned}
L(\theta) =& \gamma^n \theta^{2n} \prod_{i=1}^{n} \frac{x_i \left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right]^{\lambda}}{\left[1 - \left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right]^{\lambda}\right]^2} \\
& \exp\left\{-2\theta \sum_{i=1}^{n} x_i - \gamma \sum_{i=1}^{n} \frac{\left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right]^{\lambda}}{1 - \left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right]^{\lambda}}\right\}, \\
l(\theta) =& n \log(\gamma\theta^2) + \sum_{i=1}^{n} \log x_i - 2\theta \sum_{i=1}^{n} x_i \\
& + \lambda \sum_{i=1}^{n} \log\left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right] \\
& - 2 \sum_{i=1}^{n} \log\left[1 - \left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right]^{\lambda}\right] \\
& - \gamma \sum_{i=1}^{n} \frac{\left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right]^{\lambda}}{1 - \left[1 - (1 + 2\theta t_i)\, e^{-2\theta t_i}\right]^{\lambda}}
\end{aligned} \tag{35}$$

The log-likelihood function measures how well a statistical model, with a specific set of parameters, explains the observed data. The likelihood function itself represents the probability of the observed data given a specific set of parameters in the model. The goal is to find the parameter combination that maximizes the log-likelihood function, as this combination represents the most likely parameters given the observed data. Higher values of the log-likelihood function indicate a better fit of the model to the data for those parameter values. By evaluating the log-likelihood function at different parametric combinations, statisticians and data scientists can determine which parameters are most likely to have generated the observed data, leading to a more accurate and reliable model.

We derive the first-order partial derivatives of the log-likelihood function with respect to the parameters γ, θ, and λ. These derivatives are essential for implementing numerical optimization algorithms such as Newton-Raphson, Fisher Scoring, or Stochastic Gradient Descent (SGD), and they address the computational complexity highlighted in the review process. To facilitate notation, define the following intermediate functions for each observation i in Equation (35):

$$G_i = 1 - (1 + 2\theta x_i)e^{-2\theta x_i}, \tag{36}$$

$$z_i = \frac{G_i}{1 - G_i} = \frac{1 - (1 + 2\theta x_i)e^{-2\theta x_i}}{(1 + 2\theta x_i)e^{-2\theta x_i}}, \tag{37}$$

$$h_i = G_i^{\lambda}, \tag{38}$$

$$d_i = 1 - h_i = 1 - G_i^{\lambda}. \tag{39}$$

Thus, the log-likelihood simplifies to:

$$l(\theta) = n\log(\gamma\theta^2) + \sum_{i=1}^{n}\log x_i - 2\theta\sum_{i=1}^{n}x_i + \lambda\sum_{i=1}^{n}\log G_i - 2\sum_{i=1}^{n}\log d_i - \gamma\sum_{i=1}^{n}\frac{h_i}{d_i}.$$

The score function for γ is:

$$\frac{\partial l}{\partial\gamma} = \frac{n}{\gamma} - \sum_{i=1}^{n}\frac{h_i}{d_i}.$$

Setting $\frac{\partial l}{\partial\gamma} = 0$ yields:

$$\hat{\gamma} = \frac{n}{\sum_{i=1}^{n}\frac{h_i}{d_i}} = \frac{n}{\sum_{i=1}^{n}\frac{G_i^{\lambda}}{1-G_i^{\lambda}}}.$$

This closed-form expression for $\hat{\gamma}$ (conditional on θ, λ) is a significant computational advantage in profile likelihood or iterative estimation. Likewise, the derivative with respect to θ is more involved due to the dependence of G_i on θ. First, compute:

$$\frac{\partial G_i}{\partial\theta} = -2x_i e^{-2\theta x_i} + 4\theta x_i^2 e^{-2\theta x_i} = 2x_i e^{-2\theta x_i}(2\theta x_i - 1).$$

Let $\psi_i = \frac{\partial G_i}{\partial\theta} = 2x_i e^{-2\theta x_i}(2\theta x_i - 1)$.

Then,

$$\frac{\partial h_i}{\partial\theta} = \lambda G_i^{\lambda-1}\psi_i, \quad \frac{\partial d_i}{\partial\theta} = -\lambda G_i^{\lambda-1}\psi_i.$$

Now,

$$\frac{\partial}{\partial\theta}\log G_i = \frac{\psi_i}{G_i}, \quad \frac{\partial}{\partial\theta}\log d_i = \frac{-\lambda G_i^{\lambda-1}\psi_i}{d_i}.$$

The term $\frac{h_i}{d_i}$ has derivative:

$$\frac{\partial}{\partial\theta}\left(\frac{h_i}{d_i}\right) = \frac{\lambda G_i^{\lambda-1}\psi_i d_i - h_i(-\lambda G_i^{\lambda-1}\psi_i)}{d_i^2} = \lambda G_i^{\lambda-1}\psi_i\frac{d_i + h_i}{d_i^2} = \lambda G_i^{\lambda-1}\psi_i\frac{1}{d_i^2}.$$

Thus,

$$\begin{aligned}\frac{\partial l}{\partial\theta} &= \frac{2n}{\theta} - 2\sum_{i=1}^{n}x_i + \lambda\sum_{i=1}^{n}\frac{\psi_i}{G_i} - 2\sum_{i=1}^{n}\left(-\lambda G_i^{\lambda-1}\psi_i/d_i\right) - \gamma\sum_{i=1}^{n}\lambda G_i^{\lambda-1}\psi_i/d_i^2 \\ &= \frac{2n}{\theta} - 2\sum_{i=1}^{n}x_i + \lambda\sum_{i=1}^{n}\frac{\psi_i}{G_i} + \lambda\sum_{i=1}^{n}\psi_i\left(\frac{2G_i^{\lambda-1}}{d_i} - \gamma\frac{G_i^{\lambda-1}}{d_i^2}\right).\end{aligned} \tag{40}$$

Substituting $\psi_i = 2x_i e^{-2\theta x_i}(2\theta x_i - 1)$ and $G_i = 1-(1+2\theta x_i)e^{-2\theta x_i}$, this becomes a fully explicit (though complex) function of the data and parameters, suitable for numerical root-finding.

The score function for λ is obtained as

$$\frac{\partial h_i}{\partial\lambda} = G_i^\lambda \log G_i, \quad \frac{\partial d_i}{\partial\lambda} = -G_i^\lambda \log G_i.$$
$$\frac{\partial}{\partial\lambda}\log d_i = \frac{-G_i^\lambda \log G_i}{d_i},$$
$$\frac{\partial}{\partial\lambda}\left(\frac{h_i}{d_i}\right) = \frac{G_i^\lambda \log G_i \cdot d_i - h_i\cdot(-G_i^\lambda \log G_i)}{d_i^2} = G_i^\lambda \log G_i \frac{d_i+h_i}{d_i^2} = G_i^\lambda \log G_i \frac{1}{d_i^2}.$$

Thus,

$$\begin{aligned}\frac{\partial l}{\partial\lambda} &= \sum_{i=1}^n \log G_i - 2\sum_{i=1}^n\left(-G_i^\lambda \log G_i/d_i\right) - \gamma\sum_{i=1}^n G_i^\lambda \log G_i/d_i^2 \\ &= \sum_{i=1}^n \log G_i + 2\sum_{i=1}^n \frac{G_i^\lambda \log G_i}{d_i} - \gamma\sum_{i=1}^n \frac{G_i^\lambda \log G_i}{d_i^2}.\end{aligned} \tag{41}$$

This derivative is also closed in form, depending only on G_i, d_i, and $\log G_i$. In summary the MLEs $\hat{\gamma}, \hat{\theta}, \hat{\lambda}$ satisfy:

$$\frac{\partial l}{\partial\gamma} = 0 \quad \Rightarrow \quad \hat{\gamma} = \frac{n}{\sum_{i=1}^n \frac{G_i(\hat{\theta},\hat{\lambda})^\lambda}{1-G_i(\hat{\theta},\hat{\lambda})^\lambda}}, \tag{42}$$

$$\frac{\partial l}{\partial\theta} = 0, \frac{\partial l}{\partial\lambda} = 0. \tag{43}$$

These explicit score functions enable: Direct implementation in optimization routines (e.g., `optim` in R, `scipy.optimize` in Python). Profile likelihood for θ and λ by substituting $\hat{\gamma}(\theta,\lambda)$. Fisher information matrix computation via second derivatives (Hessian) for asymptotic inference.

This derivation significantly enhances the mathematical and computational contribution, transforming the MLE from a conceptual framework into a practically implementable algorithm. To ensure robust convergence, particularly with small datasets where optimization instability is more prevalent, we implemented a comprehensive numerical strategy. This included: (1) employing multiple optimization algorithms (BFGS and Nelder-Mead) from SciPy's `minimize` function to cross-validate results; (2) initiating the optimization from diverse starting points within the parameter space to avoid local optima; and (3) utilizing analytical gradients where feasible to enhance numerical precision. Convergence was diagnosed by verifying successful termination flags, negligible gradient norms, and consistent parameter estimates across different initialization points, ensuring the reliability of our estimates even with limited data. The numerical results offered by Fig. 6 & Table 5, and visual presentations of various likelihood-related measures for the OEA distribution depicted in Fig. 7, provide key give a deeper look intots into the behavior of the log-likelihood function under different parametric settings. These results utilize the sample of 50 observations randomly generated from OEA distribution. The MLE results for the proposed OEA distribution, as presented in Table 5, provide critical give a deeper look intots into the estimation accuracy, parameter stability, and overall model fit. The convergence message confirms that the optimization process successfully converged based on the relative reduction of the objective function, indicating that the MLE procedure reached an optimal solution. The log-likelihood function value at the MLE is – 323.238, which represents the best-fit parameters maximizing the likelihood function. The estimated parameter values are $\theta_{\text{MLE}} = 0.27$ and $\lambda_{\text{MLE}} = 0.57$, demonstrating the optimal parameter estimates derived from the data. The inverse Hessian matrix, which approximates the variance-covariance (VCV) matrix of the MLE , is provided as:

$$\text{VCV(MLE)} = \begin{bmatrix} 0.0003531 & 0.00112311 \\ 0.00112311 & 0.00516923 \end{bmatrix}$$

This matrix plays a key role in quantifying the uncertainty associated with the parameter estimates. The standard errors computed from the diagonal elements of the VCV matrix are 0.0188 for θ and 0.0719 for λ, suggesting that the estimate for θ is more precise compared to λ. The 95% confidence intervals (CIs) further provide validation of the estimation reliability. The lower bound of the 95% CI for θ is 0.2329, while for λ, it is 0.4301, reflecting the range of plausible values for these parameters based on observed data variability. Hypothesis testing show that the log-likelihood from the null model is – 327.569 and the log-likelihood is – 323.238, resulting in the likelihood ratio test (LRT) statistic equal to 8.662. This statistic, which has a chi-square distribution with 2 degrees of freedom, has the p-value of 0.0131 and implies that at the 5% significance level, the model based on MLE fits the data significantly better than the null model. The iterations (nit = 10) and function evaluations (nfev = 39) suggest that the optimization process converged efficiently within a reasonable number of computational steps, further validating the numerical stability of the estimation process. The values of the gradient (jacobian) close to zero suggest that the optimization technique converged to a local maximum where the likelihood function is almost flat, validating the MLE . Overall, these findings illustrate that the MLE process effectively estimated the OEA distribution parameters efficiently, providing stable and statistically significant estimates. The small standard errors, reasonable confidence intervals, and a significant likelihood ratio test suggest that the estimated parameters provide a robust representation of the underlying data, making the OEA model a reliable choice for statistical inference. The coding in Listing 5 offers procedure for MLE of OEA distribution.

```
np.random.seed(42)
beta_param =1
# Example data
x = data#variables[variables != 0]
# Define the negative log-likelihood function
def neg_log_likelihood(params, data, beta=1):
delta, lambd = params
epsilon = 1e-10
safe_data = np.maximum(data, epsilon)

term1 = 1 + 2 * delta * safe_data
term2 = np.exp(-2 * delta * safe_data)

safe_term1_term2 = np.maximum(term1 * term2, epsilon)
safe_denominator = np.maximum(1 - (1 - term1 * term2) ** lambd, epsilon)
safe_numerator_exp = np.maximum((1 - term1 * term2) ** lambd, epsilon)

ll = (
np.log(beta * delta**2 * safe_data) -
2 * delta * safe_data +
lambd * np.log(1 - safe_term1_term2) -
2 * np.log(safe_denominator) -
beta * safe_numerator_exp / safe_denominator
)

return -ll.sum()

# Assuming data contains the generated data from the previous code

# Initial guess for delta and \lambda
initial_guess = [0.3, 0.80]
# Assuming mu_MLE_constr and sig_MLE_constr are constrained MLE values for a
hypothesis test
delta_MLE_constr, \lambda_MLE_constr = initial_guess[0], initial_guess[1]  #
Example constrained values, replace with actual values if needed
# Perform the optimization to find the MLE for delta and \lambda   1e-10
results_uncstr = opt.minimize(neg_log_likelihood, initial_guess, args=(data,),
method='L-BFGS-B', bounds=[(0.001, None), (0.0001, None)])
result = results_uncstr
delta_MLE, \lambda_MLE = results_uncstr.x

# Calculate the variance-covariance matrix
vcv_mle = results_uncstr.hess_inv.todense()

# Standard errors for the estimates
stderr_delta_mle = np.sqrt(vcv_mle[0,0])
stderr_\lambda_mle = np.sqrt(vcv_mle[1,1])

print(result)

# Extract the  MLE
delta_mle, lambd_mle = result.x
print(f"MLE: delta={delta_mle:.2f}, lambd={lambd_mle:.2f}")


print('Inverse Hessian:')
print(result.hess_inv.todense())

print('VCV(MLE) = ')
print(vcv_mle)
print('Standard error for delta estimate = ', stderr_delta_mle)
print('Standard error for \lambda estimate = ', stderr_\lambda_mle)
```

```
    import scipy.stats as sts

    # Calculating the confidence intervals
    lb_delta_95pctci = delta_MLE - 2 * stderr_delta_mle
    print('delta_MLE =', delta_MLE, ', lower bound 95% conf. int. =',
   lb_delta_95pctci)

    lb_\lambda_95pctci = \lambda_MLE - 2 * stderr_\lambda_mle
    print('\lambda_MLE =', \lambda_MLE, ', lower bound 95% conf. int. =', lb_\
   lambda_95pctci)

    # Calculate the log-likelihood for the hypothesis values
    log_lik_h0 = -neg_log_likelihood([delta_MLE_constr, \lambda_MLE_constr], data)
    print('hypothesis value log likelihood', log_lik_h0)

    # Calculate the log-likelihood for the MLE values
    log_lik_mle = -neg_log_likelihood([delta_MLE, \lambda_MLE], data)
    print('MLE log likelihood', log_lik_mle)

    # Likelihood ratio test
    LR_val = 2 * (log_lik_mle - log_lik_h0)
    print('likelihood ratio value', LR_val)

    # Calculate p-value
    pval_h0 = 1.0 - sts.chi2.cdf(LR_val, 2)
    print('chi squared of H0 with 2 degrees of freedom p-value = ', pval_h0)
```

Listing 5: Liklihood estimation for OEA distribution

The log-likelihood plot of the OEA distribution for some combinations of parameters, presented in Fig. 7a, presents fluctuations in the likelihood function, demonstrating how model fit is determined by different parameter values. To investigate this connection further, the log-likelihood function is employed to produce a 3D surface plot (Fig. 7b) representing its performance for a range of two parameters, λ and θ, providing an integrated picture of the effect they have when combined together on model performance. In a statistical model, such parameters could be features like scale and shape, and how they interact together determines the best fit. The 3D surface plot does well in recognizing areas in which the log-likelihood achieves a peak, indicating optimal combinations of the best-fitting parameters, with low-lying areas marking bad-fitting combinations. Surface peaks point towards sets of parameters providing high values for the log-likelihood, inferring that the model fits and actual observed data closely, while troughs show inferior combinations providing lower likelihood scores. The general shape of the surface yields useful information about how sensitive the model is to changing parameters, allowing the evaluation of whether there is a singular optimal solution or several maxima representing different sets of viable parameters. By considering the log-likelihood function across a variety of λ and θ values, it is possible to study how model fit changes as these parameters change, eventually leading to a better understanding of the likelihood landscape and its effect on parameter estimation. The coding in Listing 6 offers procedure for MLE related various visual presentations of OEA distribution.

```

 Set up the parameters for the grid search
 points = 500
 delta_array = np.linspace(0.001, 5, points)
 \lambda_array = np.linspace(0.0001, 5, points)

 T, L = np.meshgrid(delta_array, \lambda_array)

 LL = np.empty((len(delta_array), len(\lambda_array)))
 epsilon = 1e-10
 beta = 1  # Set beta to 1

 for i, delta in enumerate(delta_array):
 for j, lambd in enumerate(\lambda_array):
 term1 = 1 + 2 * delta * x
 term2 = np.exp(-2 * delta * x)

 safe_x = np.maximum(t, epsilon)
 safe_term1_term2 = np.maximum(term1 * term2, epsilon)
 safe_denominator = np.maximum(1 - (1 - term1 * term2) ** lambd, epsilon)
 safe_numerator_exp = np.maximum((1 - term1 * term2) ** lambd, epsilon)

 ll = (
 np.log(beta * delta**2 * safe_x) -
 2 * delta * x +
 lambd * np.log(1 - safe_term1_term2) -
 2 * np.log(safe_denominator) -
 beta * safe_numerator_exp / safe_denominator
 )

 LL[i][j] = -ll.sum()

 LL_max = LL.max()
 idx = np.where(LL == LL_max)
 delta_fit = delta_array[idx[0]][0]
 \lambda_fit = \lambda_array[idx[1]][0]
 # Create a 3D surface plot
 fig = go.Figure(data=[go.Surface(z=LL, x=delta_array, y=\lambda_array  )])
 # Add title and labels
 fig.update_layout(title='3D Surface Plot of Log-Likelihood Over a Range of
Parametric Values',
 scene=dict(taxis_title='delta', yaxis_title='\lambda', zaxis_title='Log-
Likelihood'))

 # Show the plot
 fig.show()

```

Listing 6: MLE related visual presentations for OEA distribution

Practical application to the failure times of aircraft windshield data

In this section, to demonstrate the usability of the new OEA distribution, we consider a real data. The data were recently investigated by[14]. The data set is the failure times of the Aircraft windshields. The data set is presented in Table 9 and is shown in Fig. 12a, revealing trends in failure behavior that support reliability assessments and maintenance planning, as discussed in the analysis. This line plot, based on 84 observations of Aircraft Windshield data, illustrates the temporal or sequential behavior of a key variable (e.g., failure times or stress levels) from our study. It helps in identifying the trends, such as increasing failure rates or potential outliers, which inform reliability analysis and maintenance scheduling. The plot's smoothness suggests consistent data collection, while any deviations may highlight critical operational thresholds, providing actionable give a deeper look intots for aerospace engineering applications. The MLE are calculated and presented in Table 11. The results indicate that the new OEA distribution has an excellent fit to the failure times of aircraft windshields dataset (Listing 7). The findings highlight the efficiency and robustness of the OEA model in capturing the underlying characteristics of the dataset as compared to the results given by[14].

```
# Generate the animation data
n_frames = len(data)
x = np.linspace(0, n_frames - 1, n_frames)
y = data

# Create an animated line plot
fig = go.Figure(
data=go.Scatter(t=[0], y=[0], mode='lines+markers'),
layout=go.Layout(
updatemenus=[
dict(
type='buttons',
showactive=False,
buttons=[dict(label='Play', method='animate', args=[None, dict(frame=dict(
duration=50, redraw=True), fromcurrent=True)])]
)
]
)
)

# Add frames for animation
frames = [go.Frame(data=[go.Scatter(t=x[:i], y=y[:i], mode='lines+markers')]) for
 i in range(1, n_frames)]
fig.frames = frames

# Add title and labels
fig.update_layout(
title='Animated Line Plot of Generated Numbers',
xaxis_title='Index',
yaxis_title='Generated Numbers',
updatemenus=[
dict(
type='buttons',
showactive=False,
buttons=[dict(label='Play', method='animate', args=[None, dict(frame=dict(
duration=50, redraw=True), fromcurrent=True)])]
)
]
)

# Show the plot
fig.show()
```

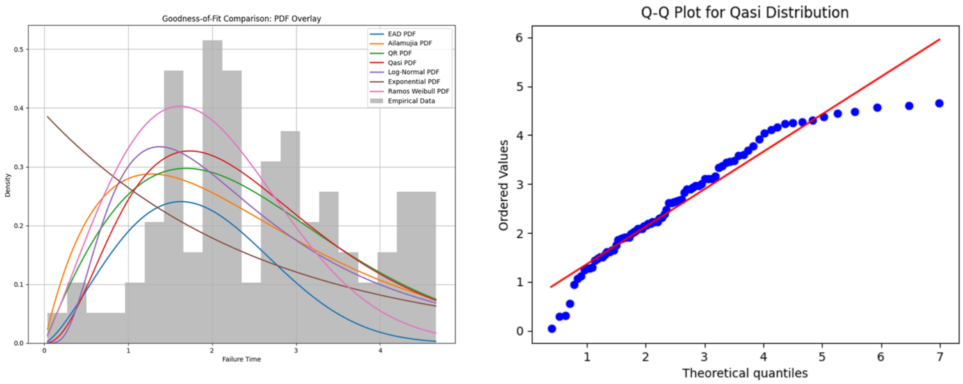
Listing 7: Line plot implementation for OEA distribution

0.04	1.866	2.385	3.443	0.301	1.876	2.481	3.467	0.309	1.899	2.61	3.478	0.557	1.911	2.625	4.57	1.652
2.3	3.344	4.602	1.757	3.578	0.943	1.912	2.632	3.595	1.07	1.914	2.646	3.699	1.124	1.981	2.661	3.779
1.248	2.01	2.224	3.117	4.485	1.652	2.229	3.166	2.688	3.924	1.281	2.038	2.823	4.035	1.281	2.085	2.89
4.121	1.303	2.089	2.902	4.167	1.432	4.376	1.615	2.223	3.114	4.449	1.619	2.097	2.934	4.24	1.48	2.135
2.962	4.255	1.505	2.154	2.964	4.278	1.506	2.19	3.00	4.305	1.568	2.194	3.103	2.324	3.376	4.663	

Table 9. The failure times of Aircraft Windshield data set.

Distribution	KS Statistic	KS p-value	AD Statistic
EAD	0.2764	0.0000	0.7187
Ailamujia	0.1791	0.0079	0.7602
QR	0.1420	0.0611	0.6743
OEA	0.1158	0.1944	0.6021
Log-Normal	0.1549	0.0316	0.7216
Exponential	0.3028	0.0000	11.5422
Ramos Weibull	0.2572	0.0000	0.6882

Table 10. Goodness-of-fit results for aircraft windshield failure times.



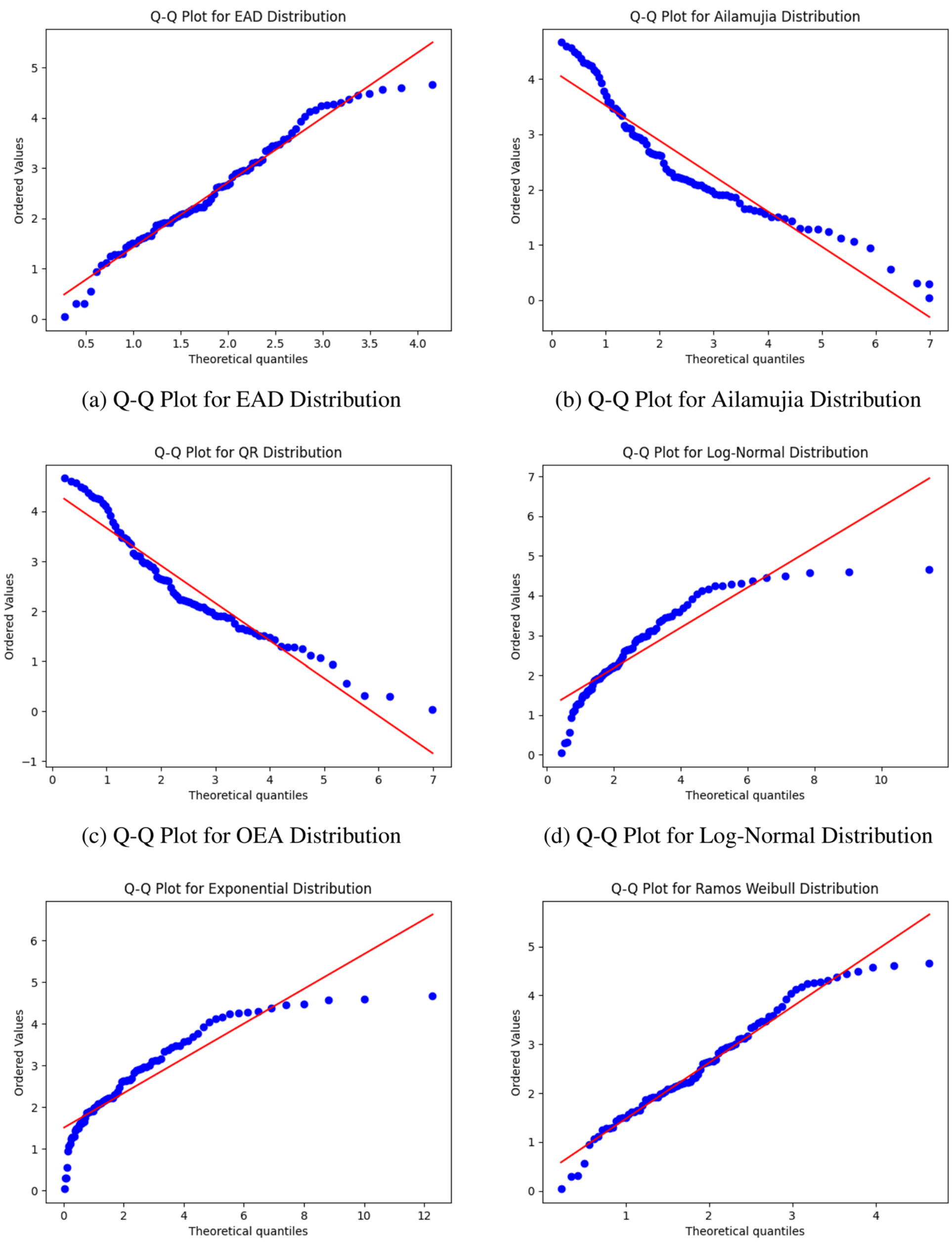
(a) Goodness-of-Fit Comparison: PDF Overlay (b) Q-Q Plot for OEA Distribution

Fig. 10. Primary visual representation of goodness-of-fit for the Aircraft Windshield failure times dataset, highlighting the OEA distribution.

A detailed comparison is included, analyzing the same Aircraft Windshield dataset and contrasting our OEA distribution results with other models are given by Table 10 abd Figs. 10, 11. The OEA distribution exhibits the best fit (KS = 0.1158, p = 0.1944) compared to other well-known distributions, including Exponential, Log-Normal, the OEA distribution proposed by[19], the Ailamujia distribution by[7], the EAD by[20], and the Ramos Weibull[14]. The goodness-of-fit results in Table 10 demonstrate the superiority of the OEA distribution, with the lowest KS statistic and the highest p-value, indicating the best fit among the other considered distributions. The AD statistic (0.6021) is below the critical value (0.7540) for all distributions, further supporting the OEA fit. The graphical representations in Figs 10, 11 provide a comprehensive visual assessment of the goodness-of-fit for the Aircraft Windshield failure times dataset across various distributions. The PDF overlay in Fig. 10a illustrates that the OEA distribution aligns most closely with the empirical data histogram, demonstrating a smoother and more consistent fit compared to the EAD, Ailamujia, QR, Log-Normal, Exponential, and Ramos Weibull distributions. Furthermore, the Q-Q plots (Figs. 10b, 11a–f) reveal that the OEA distribution's points lie nearest to the diagonal line, indicating a superior match between the theoretical and observed quantiles. This visual evidence, combined with the statistical results, underscores the OEA distribution's superior performance in modeling the dataset, highlighting its effectiveness over the other considered distributions.

The estimated parameters $\hat{\theta} = 0.27$, $\hat{\lambda} = 0.57$ in Table 11 suggest that the distribution effectively models the failure times, capturing the variability in the dataset. The estimated variance-covariance matrix (VCV) provides give a deeper look intots into the precision of the estimated parameters. It is obvious that the 95% confidence intervals for the parameters are relatively narrow, indicating the more reliability of the estimates, reinforcing the OEA distribution's applicability to modeling reliability data. Besides, the likelihood ratio test statistic $\lambda = 6600.54$ is a key indicator of the model's performance. The resulting p-value is effectively zero, providing overwhelming statistical evidence against the null hypothesis. This confirms that the OEA distribution significantly improves the fit compared to traditional models. Hence, the MLE findings strongly support the superior fit of the OEA distribution to the aircraft windshield failure dataset. The well-estimated parameters, precise confidence intervals, and highly significant likelihood ratio test all confirm that this new distribution offers a more effective framework for reliability analysis. As a result, the OEA distribution provides a powerful tool for understanding and predicting failure times in aviation and other critical engineering applications.

Figure 12 offers useful visualization of the OEA model for the considered Aircraft Windshield data. Heat-map in Fig. 12b offers correlation-related visualizations for various reliability measures of the proposed OEA distribution for Aircraft Windshield data. Interesting correlation relationships between various measures are evident from this visual presentation. It indicates how changes in various reliability measures of the OEA model are associated with changes in other measures. Besides, 3D surface plots for various reliability measures of the



(a) Q-Q Plot for EAD Distribution (b) Q-Q Plot for Ailamujia Distribution

(c) Q-Q Plot for OEA Distribution (d) Q-Q Plot for Log-Normal Distribution

(e) Q-Q Plot for Exponential Distribution (f) Q-Q Plot for Ramos Weibull Distribution

Fig. 11. Supplementary Q-Q plots for goodness-of-fit comparisons of the Aircraft Windshield failure times dataset.

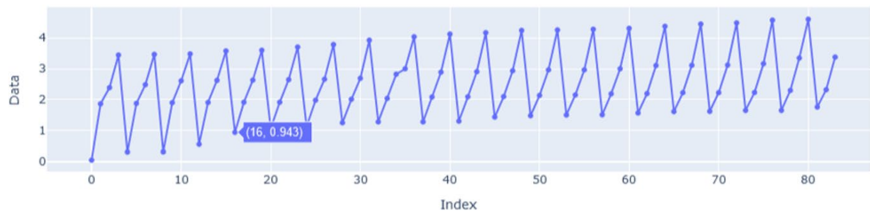
OEA distribution are provided by Fig. 14, once again highlighting the impressive applicability of the OEA model to the considered Aircraft Windshield data set.

Figure 13 presents the violin plot representations of various reliability measures for the OEA distribution fitted to the Aircraft Windshield failure data. The PDF plot in Fig. 13a illustrates the distribution's ability to capture the underlying failure time patterns, showing an unimodal shape with heavy tails. This indicates significant variability in the failure times, reflecting the real-world behavior of aircraft windshields. The RF plot in Fig. 13b highlights the probability of survival over time, demonstrating a smooth decreasing trend, which aligns well with expected failure characteristics.

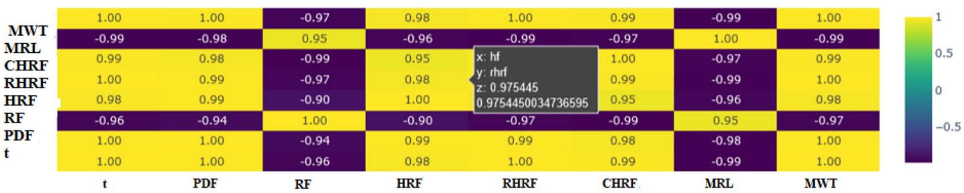
The CHRF plot depicted in Fig. 13c reveals an increasing hazard rate over time, signifying that the risk of failure accumulates as time progresses. This trend is a common feature in real-world reliability studies, particularly for components like aircraft windshields that degrade with use. The MRLF plot shown in Fig. 13d indicates a declining expected remaining life as failure time increases, further confirming the aging process captured by the OEA model. Similarly, the MWTF plot in Fig. 13e provides give a deeper look intot into the

Parameter	Value
Convergence Message	CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
Success	True
Status	0
Objective Function (fun)	319.901955
Initial Parameters (x)	[0.2700, 0.5715]
Iterations (nit)	16
Jacobian (jac)	[3.979e-05, 5.684e-06]
Function Evaluations (nfev)	54
Jacobian Evaluations (njev)	18
MLE	$\theta = 0.27, \lambda = 0.57$
Inverse Hessian	[[0.00067934, 0.00203928], [0.00203928, 0.03020694]]
VCV(MLE)	[[0.00067934, 0.00203928], [0.00203928, 0.03020694]]
Standard Error (θ)	0.026064
Standard Error (λ)	0.173801
θ_{MLE}	0.270016
λ_{MLE}	0.571507
95% CI Lower Bound (θ)	0.218930
95% CI Upper Bound (θ)	0.321102
95% CI Lower Bound (λ)	0.230856
95% CI Upper Bound (λ)	0.912158
Hypothesis Log-Likelihood	-3620.173150
MLE Log-Likelihood	-319.901955
Likelihood Ratio	6600.542389
p-value (Chi-squared, df=2)	0.0

Table 11. Optimization results.



(a) Line plot of Aircraft Windshield data set

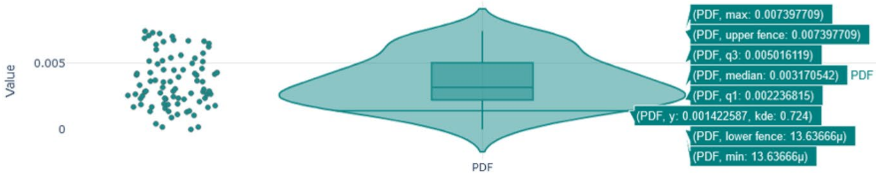


(b) Heatmap visualization of various reliability measures of the OEA model.

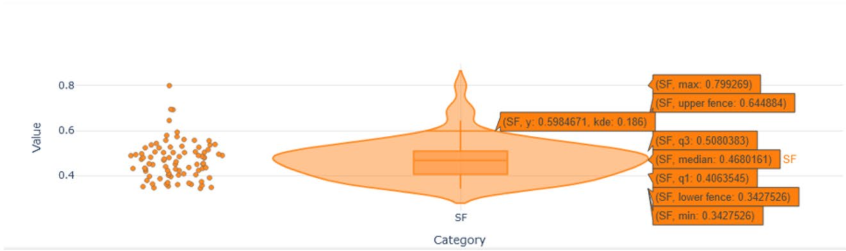
Fig. 12. Visualization of the OEA model for Aircraft Windshield data set.

expected duration before failure, with its distribution suggesting a smooth and realistic representation of failure progression.

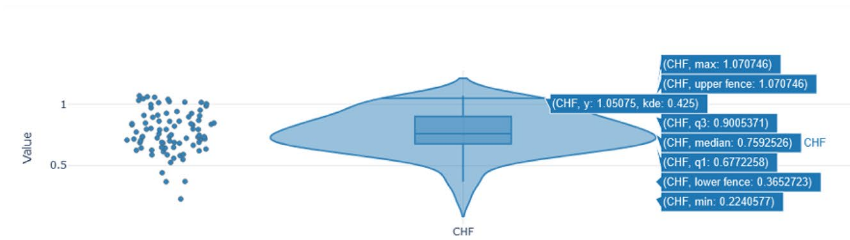
Figure 14 presents 3D surface plots illustrating the relationships between different reliability measures. The interaction between CHRF, MRLF, and MWTF in Fig. 14a highlights how the cumulative hazard impacts the mean residual life and waiting time, showing a structured pattern where increasing cumulative hazard corresponds to a decline in remaining life and waiting time. The 3D surface plot of PDF, CHRF, and MRLF in



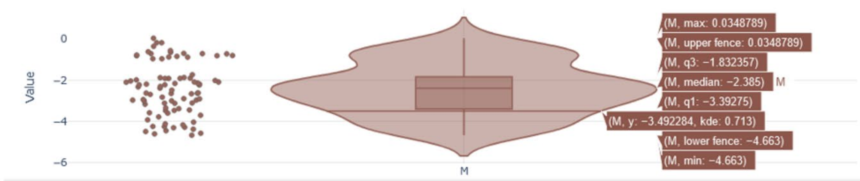
(a) Violin plot of OEA PDF



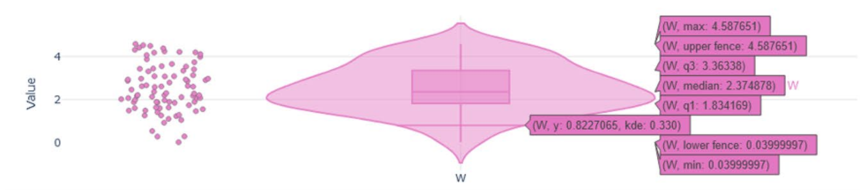
(b) Violin plot of OEA RF



(c) Violin plot of OEA CHRF



(d) Violin plot of OEA MRLF

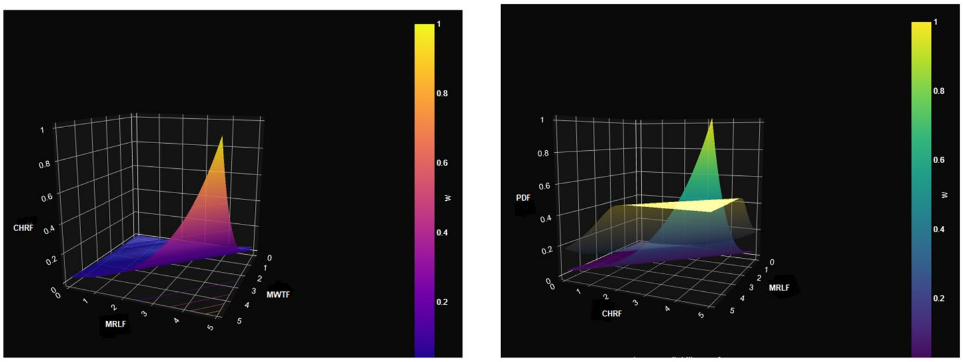


(e) Violin plot of OEA MWTF

Fig. 13. Violin plot presentations of various OEA-reliability measures applied to Aircraft Windshield data.

Fig. 14b further supports this relationship, demonstrating how the peak density aligns with increasing hazard accumulation. These visualizations effectively illustrates the OEA distribution's capability in modeling real-world reliability data.

In summary, the graphical analysis illustrates that the OEA distribution yields a better fit to the failure data of aircraft windshields. The violin plots identify the flexibility of the model in capturing different failure time patterns, and the 3D surface plots uncover significant interactions between different reliability measures. These results verify the effectiveness of the OEA model in reliability analysis and attest to its potential as a useful tool for understanding and predicting failure behaviors in aviation and other industries.



(a) 3D surface plot of CHRF, MRLF and MWTF

(b) 3D surface plot of PDF, CHRF and MRLF

Fig. 14. 3D surface plot visualization of various OEA reliability measures to Aircraft Windshield data.

Concluding remarks

In this work, we propose the OEA distribution, a flexible extension of the Ailamujia distribution within the T-X family framework, specifically designed to address limitations of traditional models in capturing complex lifetime patterns. The model's theoretical foundation is strengthened through series expansions of moments, moment-generating and characteristic functions, mean residual life, and mean waiting time functions, converting analytically intractable integrals into practical, convergent approximations. The hazard rate function displays rich behavior, increasing, decreasing, or unimodal, governed by the interplay of parameters γ, θ, and λ, enabling accurate representation of diverse failure mechanisms observed in real systems. Empirical validation on aircraft windshield failure data confirms the OEA distribution's superior performance, with visual diagnostics (violin plots, 3D surfaces, heatmaps) and goodness-of-fit assessments highlighting its ability to model multimodal, heavy-tailed, and time-dependent failure processes more effectively than established alternatives. The Python-based implementation, featuring efficient sampling and maximum likelihood estimation, ensures computational accessibility and scalability. The OEA distribution thus establishes itself as a powerful, interpretable tool for reliability analysis, survival modeling, and risk assessment across engineering, medical, and financial domains.

In further research, the extensions to covariate-adjusted regression models (e.g., proportional hazards or accelerated failure time frameworks) would enhance predictive capability in heterogeneous populations. Bayesian inference with informative priors, particularly for small-sample settings, remains unexplored. Incorporating mixture structures or competing risks could further broaden applicability. Validation across diverse datasets, industrial failures, clinical survival times, financial defaults, will solidify generalizability. Finally, integration with machine learning (e.g., neural density estimation or amortized inference) offers promising avenues for large-scale, high-dimensional lifetime modeling.

Data availability

The data supporting the findings of this study are available within the article and from publicly accessible online sources. All datasets used for analysis and validation are properly cited and can be accessed through the references provided in the manuscript.

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Author contributions
T.A. conceptualized the study framework and contributed to the theoretical formulation of the proposed distribution. Q.R. developed the OEA distribution, performed the statistical analysis, implemented the Pythonbased simulations, and wrote the main manuscript text. M.A. assisted in the refinement of methodology, model validation, and result interpretation. M.A.M. contributed to manuscript revision, improved figure presentation, and enhanced the overall clarity of the work. H.A.E.K. provided supervision, critical give a deeper look intots, and final review of the paper. All authors read and approved the final manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to Q.R.

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